

Neutral Endopeptidase-Resistant C-Type Natriuretic Peptide Variant Represents a New Therapeutic Approach for Treatment of Fibroblast Growth Factor Receptor 3–Related Dwarfism

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ABSTRACT

Achondroplasia (ACH), the most common form of human dwarfism, is caused by an activating autosomal dominant mutation in the fibroblast growth factor receptor-3 gene. Genetic overexpression of C-type natriuretic peptide (CNP), a positive regulator of endochondral bone growth, prevents dwarfism in mouse models of ACH. However, administration of exogenous CNP is compromised by its rapid clearance in vivo through receptor-mediated and proteolytic pathways. Using in vitro approaches, we developed modified variants of human CNP, resistant to proteolytic degradation by neutral endopeptidase, that retain the ability to stimulate signaling downstream of the CNP receptor, natriuretic peptide receptor B. The variants tested in vivo

demonstrated significantly longer serum half-lives than native CNP. Subcutaneous administration of one of these CNP variants (BMN 111) resulted in correction of the dwarfism phenotype in a mouse model of ACH and overgrowth of the axial and appendicular skeletons in wild-type mice without observable changes in trabecular and cortical bone architecture. Moreover, significant growth plate widening that translated into accelerated bone growth, at hemodynamically tolerable doses, was observed in juvenile cynomolgus monkeys that had received daily subcutaneous administrations of BMN 111. BMN 111 was well tolerated and represents a promising new approach for treatment of patients with ACH.

Introduction

Achondroplasia (ACH), the most common form of human dwarfism with an estimated prevalence between 1 in 16,000 to 1 in 26,000 live births (Foldynova-Trantirkova et al., 2012), is an autosomal dominant condition with the majority of new cases (80%–90%) originating de novo from parents of normal stature

(Murdoch et al., 1970; Rousseau et al., 1994). The hallmark of ACH is defective endochondral ossification, resulting in rhizomelic dwarfism, as well as skull and vertebral dysmorphism. Neurologic complications in infants due to foramen magnum stenosis and cervicomedullary compression may lead to potentially lethal hydrocephalus, hypotonia, respiratory insufficiency, apnea, cyanotic episodes, feeding problems, and quadriplegia. Mortality is increased in the first 4 years of life and in the fourth to fifth decades (Trotter and Hall, 2005; Wynn et al., 2007). Current treatments include neurosurgery and orthopedic interventions; limb lengthening to increase stature requires multiple operations over 2–3 years and remains controversial (Horton et al., 2007; Shirley and Ain, 2009). There are currently no approved pharmacologic interventions.

ACH is most commonly caused by a G380R gain-of-function mutation in the fibroblast growth factor receptor-3 (*FGFR3*) gene, resulting in sustained activation of the downstream extracellular signal-regulated kinase (ERK)/mitogen-activated protein kinase (MAPK) pathway, among others (Foldynova-Trantirkova et al., 2012), that cause supraphysiologic negative

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ABBREVIATIONS: ACH, achondroplasia; BMN 111, recombinant variant of C-type natriuretic peptide; BMN 1B2, chemically synthesized variant of C-type natriuretic peptide; BP, blood pressure; CNP, C-type natriuretic peptide; ECG, electrocardiography; ERK, extracellular signal-regulated kinase; FGF, fibroblast growth factor; FGFR3, fibroblast growth factor receptor-3; HR, heart rate; HSA, human serum albumin; IHC, immunohistochemistry; MAP, mean arterial pressure; MAPK, mitogen-activated protein kinase; MRI, magnetic resonance imaging; NEP, neutral endopeptidase; NPR, natriuretic peptide receptor; PBS, phosphate-buffered saline; PD, pharmacodynamics; PEG, polyethylene glycol; PEO, polyethylene oxide; PK, pharmacokinetics; sFGFR3, soluble fibroblast growth factor receptor-3; TDI, thanatophoric dysplasia type I; TDII, thanatophoric dysplasia type II.

regulation of chondrocyte proliferation and differentiation as well as decreased extracellular matrix synthesis (Murakami et al., 2004; Yasoda et al., 2004; Sebastian et al., 2011). In addition, stenosis of the foramen magnum and the spinal canal, caused by premature synchondrosis closure and fusion of ossification centers, is regulated by the same pathway (Hecht and Butler, 1990; Modi et al., 2008; Matsushita et al., 2009). Paracrine/autocrine factor C-type natriuretic peptide (CNP) signals through natriuretic peptide receptor (NPR) B and modulates the activity of FGFR3 through inhibition of the ERK/MAPK pathway at the level of rapidly accelerated fibrosarcoma protein kinase (RAF-1) (Krejci et al., 2005; Horton et al., 2007). CNP knockout mice, as well as those expressing mutant CNP receptors, exhibit dwarfism and have growth plates histologically similar to ACH (Rimoin et al., 1970; Naski et al., 1998; Chusho et al., 2001), whereas overexpression of CNP in mice (Kake et al., 2009) and humans (Boccardi et al., 2007; Moncla et al., 2007) is characterized by skeletal overgrowth. The dwarfism in mice overexpressing FGFR3 with a mutation analogous to human G380R (*Fgfr3*^{ACH/+}) under the control of the type II collagen promoter is corrected by endogenous CNP overproduction (Yasoda et al., 2004) or the continuous infusion of exogenous CNP (Yasoda et al., 2009), giving credence to the hypothesis that systemic administration of CNP should stimulate growth in pediatric ACH patients with open growth plates.

CNP, expressed as a 126-amino acid protein precursor (prepro-CNP), is processed to an active 53-amino acid cyclic peptide by furin and further processed to a 22-amino acid peptide by unknown proteases (Potter et al., 2006). It has been reported that only the 17-amino acid cyclic domain residues (Cys6–Cys22 of CNP22), formed by an intramolecular disulfide linkage, are required for activity (Furuya et al., 1992). Native CNP (CNP22) is rapidly cleared from the circulation by the natriuretic clearance receptor (NPR C) and neutral endopeptidase (NEP) (EC 3.4.24.11; metalloendopeptidase; enkephalinase; neprilysin; CD10, CALLA) (Brandt et al., 1995, 1997). As a result, CNP22 has a short half-life in serum of less than 2 minutes in mice and humans, thereby requiring a lengthy infusion process to result in a pharmacological benefit (Hunt et al., 1994; Yasoda et al., 2009). In fact, mice given intravenous bolus or subcutaneous administrations of CNP22 demonstrated no pharmacological benefit.

We recently described the pharmacological activity of a 39-amino acid CNP variant (BMN 111; recombinant variant of C-type natriuretic peptide), which has an extended serum half-life due to its resistance to NEP digestion (Lorget et al., 2012). We demonstrated that daily subcutaneous administrations of BMN 111 in an ACH mouse model resulted in increased axial and appendicular skeletal lengths, improvements in dwarfism-related clinical features including flattening of the skull, straightening of the tibias and femurs, and correction of the growth plate defect. Here, we report the development of BMN 111, through in vitro and in vivo approaches, which is resistant to degradation by NEP and designed to elicit the growth-promoting effects of native CNP through a subcutaneous route of administration. We also examined the cardiovascular effects of BMN 111, since it is well established that natriuretic peptides, including CNP, induce vasodilation (Clavell et al., 1993; Charles et al., 1995; Igaki et al., 1998; Scotland et al., 2005; Pagel-Langenickel et al., 2007), and then evaluated the growth-potential at doses that were considered hemodynamically acceptable [$<10\%$ drop in blood pressure (BP) and $<25\%$ increase in heart rate (HR)] in mice and monkeys. This article

focuses on the pharmacological effects of daily subcutaneous administrations of BMN 111 in mice (normal and ACH models) and normal juvenile cynomolgus monkeys.

Materials and Methods

Native CNP and Variants. Native CNP and variants were chemically synthesized using standard Fmoc chemistry (AnaSpec Inc., Fremont, CA; and GenScript USA, Inc., Piscataway, NJ). Protein sequences for coded samples were as follows: NH₂-GLSKGCFGLKLDRI SMSGLGC-COOH [native CNP; CNP22], NH₂-DLRVDTKSR-AAWARLLQEHNPARKYK GANKKGLSKGCFGLKLDRI SMSGLGC-COOH [CNP53], NH₂-GQEHNPARKYK GANKKGLSKGCFGLKLDRI SMSGLGC-COOH [BMN 1B2 chemically synthesized variant of C-type natriuretic peptide], NH₂-GHKSEVAHRFKGANKKGLSKGCFGLKLDRI SMSGLGC-COOH [chimeric peptide of human serum albumin (HSA) and CNP27; the HSA sequence (AC P02768; amino acids 27–36) is underlined] [HSA_(27–36)-CNP27], NH₂-GQEHNPARKYK GANQQGLSKGCFGLKLDRI SMSGLGC-COOH [BMN 1B2 (QQ)], NH₂-GERAFKAWAVARLSQGLSKGCFGLKLDRI SMSGLGC-COOH [chimeric peptide of HSA and CNP22; the HSA sequence (AC P02768; amino acids 231–245) is underlined] [HSA_(231–245)-CNP22], NH₂-GQPREPQVYTLPPSGLSKGCFGLKLDRI SMSGLGC-COOH [chimeric peptide of human IgG and CNP22; the IgG sequence (AC P01857; amino acids 224–237) is underlined] [IgG_(224–237)-CNP22], and NH₂-GQPREPQVYTGANQQGLSKGCFGLKLDRI SMSGLGC-COOH [chimeric peptide of human IgG and CNP27; the IgG sequence (AC P01857; amino acids 224–233) and the KK to QQ CNP27 variant sequence are underlined] [IgG_(224–233)-CNP27 (QQ)]. BMN 111 was recombinantly expressed in *Escherichia coli* (Long et al., 2012) and has the following protein sequence: NH₂-PGQEHNPARKYK GANKKGLSKGCFGLKLDRI SMSGLGC-COOH. CNP22 and all variant constructs have been oxidized to form one intramolecular disulfide bond. All peptides were $\geq 90\%$ pure and masses were confirmed by liquid chromatography/mass spectrometry.

NEP Resistance. Native CNP (CNP22) and variants (100 μ M) were incubated in the presence of purified recombinant human NEP (no. 1182-ZN-010, 1 μ g/ml; R&D Systems, Minneapolis, MN) in phosphate-buffered saline (PBS) buffer at 37°C for 140 minutes. Throughout the incubation, a portion of the sample was removed and quenched with EDTA (10 mM). Reactions were reduced with dithiothreitol (10 mM) for 30 minutes at 37°C and then analyzed by liquid chromatography/mass spectrometry. All concentrations listed are final. Results were reported as percentage of intact peptide remaining compared with time zero. All assays were repeated at least once for candidates that demonstrated native potency.

Potency (cGMP) Assay. Potency was determined in a cell-based assay using murine NIH3T3 fibroblasts, which endogenously express NPR B but not the NPR A nor NPR C receptors (Abbey and Potter, 2003). Briefly, 50%–80% confluent fibroblasts were pretreated with a phosphodiesterase inhibitor (0.75 mM isobutylmethylxanthine) in Dulbecco's modified Eagle's medium/PBS (1:1) for 15 minutes at 37°C/5% CO₂. Next, CNP22 or variants (10^{−11} M to 10^{−5} M) were added to the cells without media exchange in duplicate and incubated for an additional 15 minutes. Cells were detergent lysed (0.1% Triton X-100) and cGMP concentration was determined using a competitive immuno-based assay (CatchPoint; Molecular Devices, Sunnyvale, CA).

PEGylation. PEGylation reaction conditions were optimized to facilitate specific conjugation of polyethylene glycol (PEG) moiety at the NH₂ terminus of CNP or its variant, such as CNP27. Briefly, *N*-hydroxysuccinimide-activated PEGs of varying size (NOF America Corporation, White Plains, NY; and Thermo Fisher Scientific, Waltham, MA) were incubated with CNP22 or CNP27 at a 1:1 molar ratio in 0.1 M KPO₄ pH 6 for 1 hour at room temperature. NH₂-terminal lysines (i.e., nonring lysines) of CNP27 were changed to arginines to eliminate additional PEGylation sites without affecting NPR B binding and signaling activity (data not shown). Mono-PEGylated species were

purified by C5 reverse-phase high-performance liquid chromatography using an acetonitrile gradient containing 0.1% formic acid.

Pharmacokinetics. The pharmacokinetics (PK) profile of various CNP variants and their time courses of plasma cGMP concentrations were determined in 7- to 8-week-old male wild-type rats [Crj:CD (SD) IGS] or wild-type mice [FVB/nJ] (Charles River Laboratories, Inc., Wilmington, MA) after a single intravenous (20 nmol/kg in rats, $n = 3$; 25 nmol/kg in mice, $n = 4$) or subcutaneous (50 nmol/kg in rats, $n = 3$; 70 nmol/kg in mice, $n = 4$) administration. All peptides were formulated in 30 mM acetic acid pH 4.0 containing 10% sucrose and 1% benzyl alcohol. Plasma CNP immunoreactivity was determined using a competitive radioimmunoassay and a commercially available polyclonal antibody against the cyclic ring portion of CNP (Bachem, Bubendorf, Switzerland). Plasma cGMP concentration was determined by a competitive radioimmunoassay (Yamasa Corporation, Salem, OR).

Activity, Accumulation, and Clearance of BMN 111 at the Growth Plate. Mice were dosed and anesthetized at 15 minutes postdose, which was previously determined to coincide with the maximum cGMP response time, unless otherwise noted. Blood was collected from the heart via intracardiac puncture. Femurs with complete knee cartilage were harvested and immediately frozen in liquid nitrogen. Distal epiphysis sections from each mouse were separated for either cGMP or immunohistochemistry (IHC) experiments. For cGMP experiments, the epiphysis was pulverized using a Covaris CPO2 cryoPREP tissue homogenizer (Covaris, Inc., Woburn, MA). cGMP was extracted from the frozen pulverized epiphysis in PBS buffer containing 0.8 mM phosphodiesterase inhibitor (isobutylmethylxanthine) and quantified by competitive enzyme-linked immunosorbent assay (CatchPoint cGMP fluorescent assay kit; Molecular Devices). For IHC, tissues were fixed in 4% paraformaldehyde immediately after dissection, decalcified in 10% formic acid/PBS until no calcium oxalate precipitate formed with 5% ammonium oxalate, then dehydrated, paraffin embedded, and sectioned at 7 μ m. Sections were deparaffinized and rehydrated prior to antigen retrieval in 10 mM citrate (30 minutes, 80°C), then blocked (1% normal donkey serum, 0.1% bovine serum albumin, 0.1% NaN₃, 0.3% Triton X-100 in PBS; 1 hour, room temperature) and incubated in monoclonal CNP antibodies (4°C, overnight). Secondary donkey anti-mouse antibodies, conjugated to Alexa Fluor 488 were applied (1 hour, room temperature; Invitrogen, Carlsbad, CA). For quantification of signal intensity, confocal stacks were acquired using a Zeiss LSM 510 NLO laser scanning microscope (Carl Zeiss, Oberkochen, Germany) with a 40 $\times$ objective, 2 $\times$ zoom, and a 0.53- μ m z increment were used for IHC experiments. All experiments were performed in duplicate ($n = 2$).

Dose Regimen. Three-week-old wild-type (FVB/nJ; Charles River Laboratories, Inc.) male mice were given subcutaneous injections of BMN 111 (20 nmol/kg) daily on alternating weeks (weeks 1, 3, and 5) or vehicle [30 mM acetic acid pH 4.0 containing 10% sucrose (w/v) and 1% (w/v) benzyl alcohol] daily for 5 weeks ($n = 10$ /group). Tail measurements were collected at study initiation. Growth was monitored during the in-life treatment period by weekly tail measurements. At necropsy, final X-ray and naso-anal and tail measurements were obtained. Long bones were collected and measured for length, and the femur and tibia were fixed for histology and archived.

Pharmacological Effects of CNP Variants in Wild-Type Mice. FVB/nJ wild-type mice (Charles River Laboratories, Inc.) were administered daily subcutaneous injections at varying dose levels (20–200 nmol/kg; $n = 8$ /group) over 35 days. All CNP variants were formulated in vehicle [30 mM acetic acid buffer solution pH 4.0, containing 10% (w/v) sucrose and 1% (w/v) benzyl alcohol]. Mice (± 1 S.D. of the average body weight) were randomized at 3 weeks ± 2 days of age. Doses were given at approximately the same time each day, 2 hours prior to the dark cycle, and were based on the most recently collected body weight. The lengths of the tibia, femur, humerus, ulna, and lumbar vertebra 5 were measured with a caliper. Treated groups were compared with the vehicle control group at common time points by analysis of variance with a post hoc Dunnett's t test (Dunnett and Crisafio, 1955) or other appropriate test.

Pharmacological Effects of BMN 111 in *Fgfr3*^{ACH/+} Mice. *Fgfr3*^{ACH/+} mice were kindly provided by David M. Ornitz (Washington University in St. Louis, St. Louis, MO) and bred at Jackson Laboratories (West Sacramento, CA). Expression of activated FGFR3 was targeted to growth plate cartilage using regulatory elements from the collagen 2 gene (Naski et al., 1998). Three-week-old *Fgfr3*^{ACH/+} male mice (FVB/nJ. *Fgfr3*^{ACH/+} JAX West; $n = 8$ /group) were administered daily subcutaneous injections over 35 days (5, 20, and 70 nmol/kg). *Fgfr3*^{ACH/+} mice and their wild-type littermates were anesthetized and randomized by body weight into treatment groups. Prior to the study, mice were monitored for body weight, general health, and tail length. On day 37, all mice were euthanized by terminal anesthesia. Left and right tibia, femur, humerus, and ulna were collected and measured using a digital caliper. The left bones were fixed in 10% neutral-buffered formalin overnight, and then stored in ethanol at 2–8°C.

Hemodynamic Effects of CNP Variants in Wild-Type Mice.

Mouse studies were performed at LAB Research, Inc. (Dorval, QC, Canada). An isoflurane gas-anesthetized mouse model was used to reduce background variability in hemodynamic readouts, and to provide greater sensitivity to reduction in BP by blunting the compensatory increase in HR. CNP variants were tested over a dose range of 20–200 nmol/kg (2000 nmol/kg additional dose for BMN 111). Mice (6- to 7-week-old FVB/nJ; Charles River Laboratories, Saint-Constant, QC, Canada) were anesthetized with isoflurane gas. A pressure-monitoring catheter connected to a telemetry transmitter (PA-C10 or PXT-C50; Data Sciences International, New Brighton, MN) was placed in the aorta for arterial BP measurements. The position of the catheter was confirmed by analysis of pressure tracings. Hemodynamic parameters were recorded continuously, and were allowed to stabilize for at least 15 minutes prior to subcutaneous administration of CNP variants or vehicle control. At least 30 minutes were allowed to elapse before administration of successive doses. The mean of parameter values in the 15 minutes before dosing was compared with the mean of parameter values in the 15 minutes immediately after dosing ($n = 3$ –5/group).

Hemodynamic Effects of BMN 111 in Cynomolgus Monkeys.

All nonhuman primate studies were performed at LAB Research, Inc. At least 2 weeks prior to experimentation, animals were implanted with a cardiovascular transmitter (Data Sciences International) by which electrocardiography (ECG) data and systolic BP, diastolic BP, and mean arterial pressure (MAP) were recorded continuously via telemetry (Dataquest A.R.T.; Data Sciences International). Experiments were conducted first in isoflurane gas-anesthetized monkeys to establish the hemodynamically active dose range of BMN 111 (doses tested ranged from 0.35 to 17 nmol/kg). After anesthetic induction, hemodynamic and ECG readouts were allowed to stabilize for at least 15 minutes before administration of BMN 111. The hemodynamically active dose range was then confirmed in conscious animals (7–35 nmol/kg). In conscious monkeys, to minimize derangement of HR and BP due to animal handling, BMN 111 was administered via a long subcutaneously implanted catheter, which allowed “remote” administration without removing the animal from the cage. Mean HR and MAP values from 10 to 20 minutes postdose (covering the time of BP nadir) were compared with mean values in the 15 minutes just prior to dosing. ECG data were evaluated from 15 minutes prior to each dose through 60 minutes postdose ($n = 1$ –4/group).

Pharmacological Effects of BMN 111 in Cynomolgus Monkeys.

The effect of BMN 111 on growth was investigated in normal, growing, juvenile male cynomolgus monkeys (aged 2–4 years at the onset of treatment; 2.2–2.9 kg body weight). BMN 111 (either 2.25 or 8.25 nmol/kg, or vehicle control; $n = 4$ /group) was administered by daily subcutaneous injection for 181 days. Throughout the study, animals were monitored for mortality and clinical signs. Hematology and clinical chemistry parameters were measured on days –7, –1, 7, 21, 35, 49, 63, 77, 91, 105, 133, 161, and 182. Total serum alkaline phosphatase was measured on an automated chemistry analyzer (CiToxLAB, Laval, QC, Canada). Bone-specific alkaline phosphatase was measured using the Ostase BAP assay (Immunodiagnostic Systems Inc., Gaithersburg, MD). During pretreatment and weeks 4, 8, 13, and 23, assessments of

tibial length and growth plate width were made by digital radiographs, proximal tibial growth plate volume and width were evaluated by magnetic resonance imaging (MRI), and lengths of limbs and tail were measured with a tape measure. One day after their last BMN 111 dose, the animals were euthanized and subjected to necropsy.

MRI. Sagittal, three-dimensional, fat-suppressed spoiled gradient recalled echo imaging sequences of each knee were acquired with an eight-channel knee coil, using a high-resolution 1.5 Tesla system (GE HDx, Mississauga, ON, Canada). Sequence parameters were as follows: echo time, 15 milliseconds; repetition time, 47 milliseconds; number of averages, 3; slice thickness/gap, 1.5 mm/0; matrix, 512×512 ; and field of view, 10 cm. All measures were performed by the same veterinary radiologist. The maximal height of the proximal physis of the right and left tibia was measured in its central third using OsiriX 3.7.1 software (Pixmeo, Geneva, Switzerland). Using the brush selection tool available in the software, the surface of the physis of the right proximal tibia was then selected to include the hyperintense layer between the adjacent hypointense bone. This was repeated on all consecutive images on which the growth plate was well demarcated and surrounded with hypointense layers of bone. This technique aimed to only select the plate itself and exclude the peripheral cartilage. To avoid inclusion of this peripheral cartilage, the selection solely included the portions of the plate that presented parallel borders and excluded more peripheral portions that presented diverging margins. When the surface of the growth plate was selected on all consecutive images, its volume was calculated using the automated volume calculation plug-in included in the software ($n = 4/\text{group}$).

Radiographic Evaluation of Tibial Length. Posteroanterior projections collimated to include each of the lower limbs and centered on the knees were performed with digital computed radiography (Agfa CR-DX, Toronto, ON, Canada) and taken while the animals were under general anesthesia. Mediolateral projections of the right tibia, centered on the proximal tibial physis, were also performed. Right tibial lengths (in millimeters) were measured manually on posterior–anterior projections with dedicated image analysis software (OsiriX 3.7.1; Pixmeo). The system was calibrated and the monkey legs were placed directly on the phosphorus plates to limit magnification effects. All images were interpreted and measured by the same veterinary radiologist who remained blinded to the treatment groups ($n = 4/\text{group}$).

Postmortem Microcomputed Tomography of Lumbar Vertebrae. Lumbar vertebrae 2, 3, and 4 were excised at necropsy, fixed in formalin, and scanned using the SkyScan 1176 microCT instrument (Micro Photonics, Inc., Allentown, PA), at a resolution of $35 \mu\text{m}$, with the X-ray source set to 80 kV, 300 μA , and using a Cu + Al filter. Images were reconstructed by NRecon (Bruker MicroCT, Kontich, Belgium). To measure the foramen area of each vertebra, images were processed using the SkyScan-associated Data Viewer and the bone position was optimized. For each vertebra, the area was computed from the transaxial image corresponding to the narrowest part of the foramen in the coronal aspect ($n = 4/\text{group}$). The relevant transaxial image was saved as a single image and the foramen area measured using CTan software (Bruker MicroCT).

Histomorphometric Analysis of the Growth Plate in Cynomolgus Monkeys. For dynamic histomorphometry, calcein (10 mg/kg) was administered 14 days prior to necropsy, and oxytetracycline (40 mg/kg) was administered 6 days prior to necropsy. Left tibias were dissected, formalin fixed, dehydrated, and embedded in methyl methacrylate. Five $7\text{-}\mu\text{m}$ sections were obtained from the 50% level of the bone for analysis of the proximal growth plates and trabecular bone. Sections were stained with von Kossa, Goldner trichrome, and tartrate-resistant acid phosphatase stain ($n = 4/\text{group}$). Rate of growth was determined from the slope of length measurements plotted over time, and from fluorescent labeling of new bone.

Histomorphometric Analysis of the Bone of Cynomolgus Monkeys Treated with BMN 111. Left tibias, with growth plates intact, were harvested at necropsy, formalin fixed, and stored in 70% ethanol. Tibias were trimmed, dehydrated, and embedded in methyl methacrylate for plastic histology. Five $7\text{-}\mu\text{m}$ sections were obtained

from the 50% level of the bone for analysis of the proximal growth plates and trabecular bone. Tibias were stained with von Kossa, Goldner trichrome, and tartrate-resistant acid phosphatase stains. The combination of these three stains allowed analysis of the growth plate morphology, trabecular bone volume and architecture, quantification of unmineralized matrix (osteoid), and quantification of osteoblast and osteoclast numbers. Unstained sections were mounted for visualization of fluorescent labels for dynamic histomorphometry. Two operators measured total growth plate thickness of the right proximal tibial plate at six randomly chosen spots; 12 measurements were thereafter averaged for each sample. For each of the 12 fields, three columns of proliferating cells were assessed to determine the average number of proliferating cells per proliferating column. In addition, four regions of cuboidal chondrocytes in each field were assessed for mean cell volume of hypertrophic chondrocytes. For assessment of proliferating zone thickness and hypertrophic zone thickness, five measurements were made and averaged for each sample. Trabecular bone histomorphometry was evaluated within two $3500 \mu\text{m} \times 3500 \mu\text{m}$ regions of interest by two operators.

All procedures described herein were conducted in accordance with the principles and procedures of the National Institutes of Health Guide for the Care and Use of Laboratory Animals. Mice and rats were humanely euthanized via anesthesia with carbon dioxide (performed in accordance with accepted American Veterinary Medical Association Guidelines on Euthanasia, June 2007). The monkeys were sedated with a combination of ketamine hydrochloride and acepromazine given intramuscularly, followed by an overdose of sodium pentobarbital, followed by exsanguination.

Results

Rational Design and In Vitro Screening of Potential NEP-Resistant CNP Variants. Watanabe et al. (1997) reported that proteolysis of CNP22 by NEP occurred after initial attack at the Cys6–Phe7 bond. To test this, we synthesized peptidomimetics of CNP22 that contained either a reduced or methylated amide bond between Cys6 and Phe7 (Cys-methylene and *N*-methyl-Phe7, respectively) of CNP22 and incubated in the presence of purified human NEP. Analysis of the digestion products revealed that the Cys–Phe peptidomimetic bond was resistant to NEP in both variants (data not shown). However, when measuring the rate of disappearance of the intact molecule, these variants were indistinguishable from CNP22, indicating that proteolysis occurred at other sites of CNP22 and does not depend on initial cleavage of the Cys–Phe bond (Table 1).

Oefner et al. (2000) proposed that the size-limited active site cavity of NEP restricts substrates based on their size ($<3 \text{ kDa}$), a claim that is supported by natural substrate data (Kerr and Kenny, 1974; Erdös and Skidgel, 1989; Vijayaraghavan et al., 1990). To test this, we made larger variants of CNP through PEG conjugation, native CNP amino acid extensions, or by fusing CNP to other peptide sequences (chimeras). CNP variants, produced by chemically conjugating PEG units to the peptide NH_2 terminus, exhibited size-dependent resistance to NEP proteolysis. Specifically, NEP resistance was observed in PEGylated CNP22 variants in which the molecular mass of the PEG unit was $\geq 1 \text{ kDa}$ or when the total molecular mass of the PEG–CNP22 conjugate exceeded 3.2 kDa . However, these PEG–CNP conjugates were poor agonists of NPR B (≥ 16 -fold increase in EC_{50}). Interestingly, PEGylation of a longer native CNP sequence (CNP27) reclaimed the lost potency, while maintaining NEP resistance (Table 1).

Similar size-dependent results were observed by increasing the size of CNP22 through amino acid extensions. Here, we

TABLE 1

In vitro potency and NEP resistance for CNP variants

Data are expressed $\pm$ S.D. values as applicable.

Description	Molecular Mass	Potency ^a (EC ₅₀)	NEP Resistance ^b
	<i>kDa</i>	<i>nM</i>	% Intact
CNP22	2.2	13 $\pm$ 5.4	2.4 $\pm$ 1.8
CNP22, K4R ^c	2.2	12 $\pm$ 1.4	<5
CNP27, K4R, K5R, K9R ^c	2.8	8.7 $\pm$ 1.8	<5
ANP28	3.1	>2000	NT
CNP22, Cys6-methylene ^d	2.2	44 $\pm$ 6.2	<5
CNP22, N-methyl-Phe7 ^d	2.2	860 $\pm$ 380	<5
CNP22, 20 kDa PEG	22	>2000	100
CNP22, 5 kDa PEG	7.2	>2000	84
CNP22, 2 kDa PEG	4.2	>2000	100
CNP22, 1 kDa PEO24	3.2	640 $\pm$ 320	90
CNP22, 0.6 kDa PEO12	2.8	210 $\pm$ 30	40
CNP27, 2 kDa PEG	4.8	>2000	100
CNP27, 1 kDa PEO24	3.8	16 $\pm$ 2.8	103 $\pm$ 2.7
CNP27, 0.6 kDa PEO12	3.4	7.8 $\pm$ 1.4	69 $\pm$ 1.6
CNP30	3.1	8.4 $\pm$ 3.9	36 $\pm$ 1.9
CNP33	3.5	11 $\pm$ 0.1	99 $\pm$ 1.2
CNP36	3.8	5.8 $\pm$ 3.5	98 $\pm$ 2.2
CNP37	3.9	11 $\pm$ 2.0	97 $\pm$ 8.3
CNP38	4.1	6.8 $\pm$ 0.4	105 $\pm$ 7.7
CNP39	4.2	17 $\pm$ 1.6	95 $\pm$ 6.8
CNP40	4.3	10 $\pm$ 2.6	101 $\pm$ 6.3
CNP53	5.8	7.1 $\pm$ 0.5	106 $\pm$ 20
BMN 1B2	4.0	8.7 $\pm$ 0.5	110 $\pm$ 0.02
BMN 1B2(QQ)	4.0	130 $\pm$ 20	102
BMN 111 ^e	4.1	4.9 $\pm$ 1.5	99 $\pm$ 0.6
HSA _(231–245) -CNP22 ^f	3.9	11 $\pm$ 3.2	20 $\pm$ 0.6
IgG1 _(224–237) -CNP22 ^f	3.7	72 $\pm$ 5.9	75
IgG1 _(224–233) -CNP27(QQ) ^f	3.9	920 $\pm$ 50	40
HSA _(27–36) -CNP27 ^f	4.0	6.9 $\pm$ 2.1	105 $\pm$ 6.4

NT, not tested.

^aMean EC₅₀ ($n \geq 2$) of cGMP production in murine NIH3T3 fibroblasts after 15-minute exposure to CNP variants (10^{-10} M to 10^{-5} M), with nonlinear curve fit using the Hill equation (Erithacus Software).^bNEP resistance was determined by measuring the amount of intact peptide remaining after exposure to human NEP for 140 minutes in PBS at 37°C ($n = 2$, for variants with near native potency; $n = 1$ for all other variants). Peptide digests were analyzed by liquid chromatography/mass spectrometry.^cPeptides used for PEGylation variants.^dNon-native Cys6-Phe7 peptide bond analogs were synthesized based on reported initial NEP cleavage site (Watanabe et al., 1997).^eBiologic synthesis (all other analogs in this table were prepared by chemical synthesis).^fChimeric sequences were synthesized on the amino terminus of CNP (IgG, Ac P01857, PDB ID 2IWG; HSA Ac P02768, PDB ID 1BM0).

synthesized native amino acids on the NH₂ terminus of CNP22 based on CNP53 sequence (active tissue expressed form of CNP). NEP resistance was observed when the total number of residues was ≥ 33 amino acids (>3.4 kDa) and all retained CNP22 potency (EC₅₀ = 7–18 nM). Variants of CNP37 designed for enhanced serum stability, BMN 1B2 (approximately 4.0 kDa) and BMN 111 (approximately 4.1 kDa), also demonstrated NEP resistance and equivalent potency to CNP22. However, glutamine substitutions at the native processing site [Lys30 to Lys31 of CNP53, BMN 1B2(QQ)], designed to mitigate generation of CNP22 after parenteral injection, were 10-fold less potent than CNP22 (EC₅₀ = 130 nM; Table 1).

Finally, to address the potential proteolytic vulnerability issues of native CNP sequence to unknown protease(s) in vivo, we designed a variety of chimeric CNP variants derived from short sequences of albumin and IgG. Non-native CNP sequences were chemically synthesized to the NH₂ terminus of CNP22 and CNP27 and were selected based on their homology between species ($>70\%$), abundance in serum (>1 mg/ml), and exposure to solvent ($>90\%$) using crystal structure data (PDB IDs 1BM0 and 2IWG). In silico database programs (<http://www.syfpeithi.de> and <http://www.imtech.res.in/raghava/hlapred>) were used to

avoid introducing human leukocyte antigen binding sites at the chimeric junction to reduce the potential of an immunogenic response. Of the four chimeras made, three were sensitive to NEP, despite having molecular masses ≥ 3.7 kDa (Table 1). This suggested that structural components may also influence NEP resistance, since native sequence constructs smaller in size (3.4 kDa) were completely resistant to NEP. Moreover, divergence away from native CNP sequence resulted in a significant decrease in potency. One chimeric variant, HSA_(27–36)-CNP27, demonstrated NEP resistance and equivalent potency to CNP22. In vitro NEP resistance and potency profiles of five CNP variants chosen for further in vivo evaluation are shown in Fig. 1.

NEP-Resistant CNP Variants Exhibit Longer Serum Half-Lives than Native CNP. NEP-resistant variants demonstrated an increase in serum half-life (approximately 7- to 16-fold after intravenous administration and 2- to 7-fold after subcutaneous administration in rats and mice) compared to CNP22, with half-life ($t_{1/2}$) = 14–23 minutes for NEP-resistant variants versus ≤ 2 minutes for CNP22 when dosed intravenously and $t_{1/2}$ = 12–25 minutes for NEP-resistant variants versus 3–5 minutes for CNP22 when dosed subcutaneously (Fig. 2; Table 2). The PK profiles were similar for

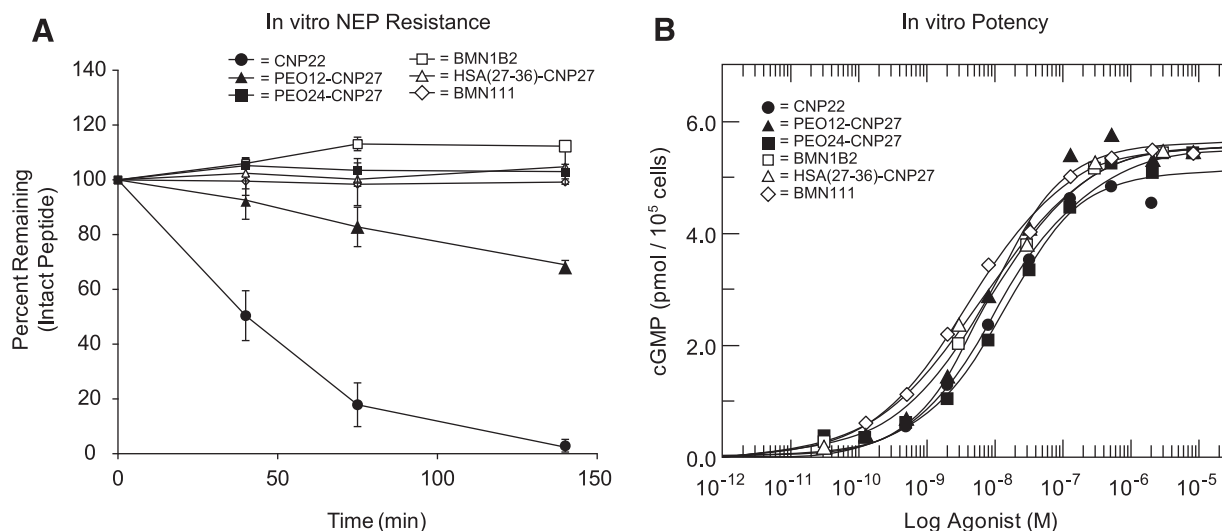


Fig. 1. NEP resistance and potency of CNP variants. (A) Plot of intact CNP22 and its variants remaining after incubation with recombinant human NEP in PBS at 37°C for 140 minutes. The mean percentage of intact peptide remaining was determined at designated times by liquid chromatography/mass spectrometry ($n = 2$). Error bars indicate the S.D. (B) Mean cGMP production in murine fibroblasts (NIH3T3; $n \geq 2$; error bars omitted for comparison clarity). The EC_{50} value was determined after 15-minute exposure to CNP22 or variants (10^{-11} M to 10^{-5} M) using a nonlinear curve fit (Hill equation; Erithacus Software Ltd., Surrey, UK). Symbols indicate the following: CNP22 (solid circle), PEO24-CNP27 (solid square), PEO12-CNP27 (solid triangle), BMN 1B2 (open square), BMN 111 (open diamond), and HSA₍₂₇₋₃₆₎-CNP27 (open triangle).

most NEP-resistant variants tested, with the exception that the polyethylene oxide (PEO) PEGylated variant (PEO24-CNP27) demonstrated a 3-fold longer serum half-life than the other variants after subcutaneous administration (Fig. 2B). Plasma cGMP profiles, a pharmacodynamics (PD) marker of NPR B activation (Wielinga et al., 2003), correlated well with the PK profiles of the CNP variants, demonstrating a clear PK/PD relationship (Fig. 2, C and D). Interestingly, cGMP concentration is not maintained for the PEGylated variant, despite the elevated exposure of this variant at the later time points (60–180 minutes; Fig. 2D). This could be caused by receptor desensitization of NPR B, which is known to occur upon prolonged exposure to CNP (Potter and Hunter, 2001). BMN 111, the recombinant version of BMN 1B2 containing one additional proline residue at the amino terminus, also demonstrated a prolonged half-life compared with CNP22 in wild-type murine studies (Fig. 2, E and F). Importantly, our data are consistent with a model whereby NEP functions as one of the major clearance pathways of CNP and supports our hypothesis that NEP-resistant variants should have longer serum half-lives.

CNP Variant Selection Based on Stimulation of Bone Growth and Hemodynamic Effects in Wild-Type Mice. Studies in rat chondrocytes using the method developed by Krejci et al. (2005) indicated that daily 1-hour exposure to CNP22 significantly reversed the growth arrest induced by FGFR3 activation, comparable to cells continuously exposed to CNP22; these results support daily administration of CNP variants in wild-type mice (data not shown). Although PEO24-CNP27 demonstrated a superior PK profile, it failed to provide a significant growth benefit in wild-type mice compared with the placebo control in preliminary range-finding studies (data not shown). For this reason, we decided to evaluate a smaller, more potent PEG variant, PEO12-CNP27, in the comparative study.

Three-week-old wild-type FVB/nJ male mice ($n = 3$ –9/group) were given daily subcutaneous injections of CNP variants BMN 1B2, BMN 111, PEO12-CNP27, or HSA₍₂₇₋₃₆₎-CNP27 at 20, 70, or 200 nmol/kg or vehicle for 36 days. The growth of the

appendicular and axial skeletons was dose related for most of the variants tested; however, growth effects were more pronounced in mice treated with BMN 111 (Fig. 3). A significant increase in naso-anal length was detected as early as 8 days after the start of BMN 111 treatment (data not shown). The PEGylated CNP variant, PEO12-CNP27, was the least pharmacologically active of the variants tested, potentially due to poor tissue bioavailability associated with PEGylated proteins (Veronese and Pasut, 2005; Ryan et al., 2008), and performed similarly to PEO24-CNP27 in our preliminary range-finding study. After 2 weeks of treatment, axial growth (naso-anal and tail length) was evident in mice treated with the chimeric CNP variant; however, the response was not sustained beyond 3 weeks (data not shown).

Additional studies designed to look at accumulation and clearance of BMN 111 at the growth plate demonstrated that consecutive daily administrations of BMN 111 augmented the cGMP levels in the distal femur growth plate, but not in the kidney, 15 minutes after the last injection (Fig. 4A). Consistent with this augmented cGMP response, immunoreactive CNP persists for several days after the last injection in wild-type mice (Fig. 4B, right). However, the accumulated BMN 111 appears to be inactive because the cGMP response was reduced to background levels by 24 hours after administration (Fig. 4B, left). On the basis of the augmented activity response we observed after consecutive daily administrations (Fig. 4A), it is unlikely that the immunoreactive BMN 111 has caused receptor desensitization. Rather, it is more likely that BMN 111 has been inactivated through a proteolytic event. In agreement with these data, *in vivo* dose regimen studies in wild-type mice demonstrated that accelerated growth was observed only during the week when mice received daily dosing. Discontinuation of treatment at 1-week intervals resulted in a return to normal growth rate (Fig. 5).

CNP produces hemodynamic effects in mice (Lopez et al., 1997), nonhuman primates (Seymour et al., 1996), rats, dogs, and humans (Barr et al., 1996); therefore, we decided to examine

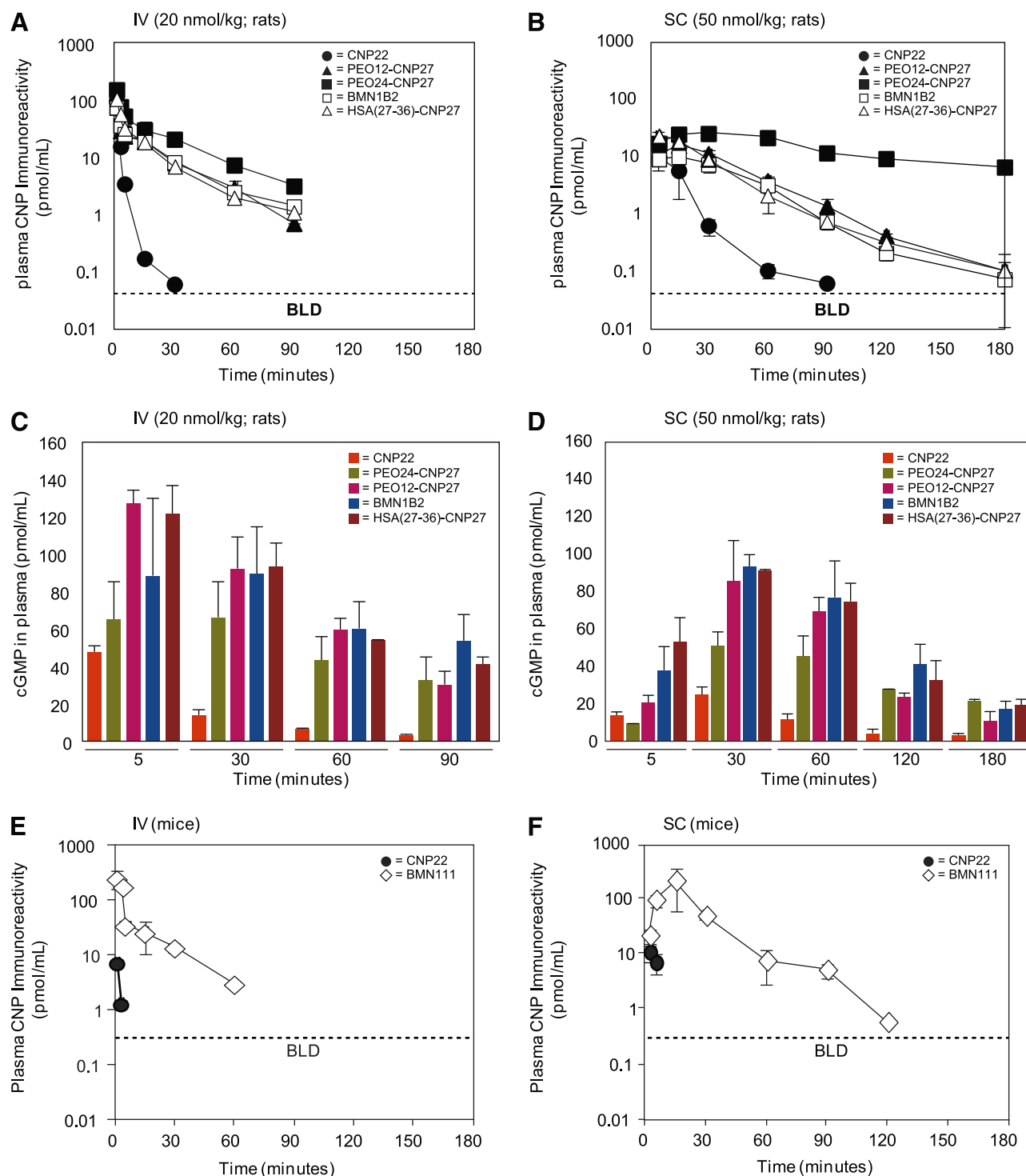


Fig. 2. PK and PD evaluation of NEP-resistant CNP variants. (A and B) Plasma CNP levels after a single intravenous (20 nmol/kg) or subcutaneous (50 nmol/kg) administration of CNP22 or variants in normal rats ($n = 3$). CNP immunoreactivity was determined using an anti-CNP rabbit polyclonal antibody in a competitive radioimmunoassay. (C and D) Plasma cGMP concentration in response to CNP binding to NPR B. cGMP concentration was determined by radioimmunoassay ($n = 3$). (E and F) Plasma CNP levels after a single intravenous (50 or 25 nmol/kg) or subcutaneous (130 or 70 nmol/kg) administration of CNP22 or BMN 111 in normal mice ($n = 4$). Error bars indicate the S.D. Symbols indicate the following: CNP22 (solid circle), PEO24-CNP27 (solid square), PEO12-CNP27 (solid triangle), BMN1B2 (open square), BMN111 (open diamond), and HSA₍₂₇₋₃₆₎-CNP27 (open triangle). BLD, below limit of detection; IV, intravenous; SC, subcutaneous.

cardiovascular effects of the CNP variants (20–200 nmol/kg) in anesthetized wild-type FVB/nJ male mice fitted with a pressure-monitoring catheter connected to a telemetry transmitter. All variants showed similar BP-reducing and HR-increasing

activity (Fig. 6). In most animals, effects were observed within 5 minutes of subcutaneous administration, with maximal drop in MAP occurring between 5 and 20 minutes postdose. This timing correlated well with the maximum concentration of the CNP

TABLE 2

PK parameters of CNP variants in wild-type rats [Crj:CD (SD) IGS] and wild-type mice [FVB/nJ]
Data are expressed with the S.D. in parentheses if applicable.

Group	Animal	Dose	Route	C_{max}	t_{max}	$t_{1/2}$	Bioavailability
		nmol/kg		pmol/ml	min	min	%
CNP22	Rat	20	i.v.	NA	NA	1.4 (0.5)	NA
PEO24-CNP27	Rat	20	i.v.	NA	NA	22 (1.5)	NA
PEO12-CNP27	Rat	20	i.v.	NA	NA	17 (1.3)	NA
BMN 1B2	Rat	20	i.v.	NA	NA	23 (3.4)	NA
HSA ₍₂₇₋₃₆₎ -CNP27	Rat	20	i.v.	NA	NA	23 (1.1)	NA
CNP22	Mice	50	i.v.	7.3 (1.1)	1 (0)	≤2	NA
BMN 111	Mice	25	i.v.	250 (86)	1.5 (1)	14	NA
CNP22	Rat	50	s.c.	9.0 (3.7)	5.0 (0.0)	10 (5.0)	19 (9.0)
PEO24-CNP27	Rat	50	s.c.	24 (1.9)	25 (8.7)	78 (16)	60 (6.0)
PEO12-CNP27	Rat	50	s.c.	15 (1.8)	12 (5.8)	25 (4.4)	24 (1.0)
BMN 1B2	Rat	50	s.c.	9.4 (2.2)	12 (5.8)	19 (4.3)	19 (4.0)
HSA ₍₂₇₋₃₆₎ -CNP27	Rat	50	s.c.	22 (4.4)	5.0 (0.0)	25 (8.5)	25 (3.0)
CNP22	Mice	130	s.c.	10 (3.2)	2.8 (1.5)	≤5	100
BMN 111	Mice	70	s.c.	200 (140)	13 (5)	15	98

NA, not available.

variants, and demonstrated a clear PK/PD relationship for this physiologic response. Because the hemodynamic responses were similar between the doses and variants tested, cardiovascular activity was not a differentiating property and no further experiments or statistical analyses were performed.

BMN 111 demonstrated an increased pharmacological activity compared with the PEGylated and chimeric CNP variants in wild-type mice, whereas the transient hemodynamic

response was very similar for the non-PEGylated CNP variants. Histomorphometric analysis of long bones showed no observable changes in trabecular and cortical architecture associated with the 5-week daily treatment of BMN 111 (data not shown), indicating that although longitudinal growth was stimulated, de novo bone formation was unaffected and normal. Based on potency and similarity to native sequence, BMN 111 was selected for studies in ACH mice and cynomolgus monkeys.

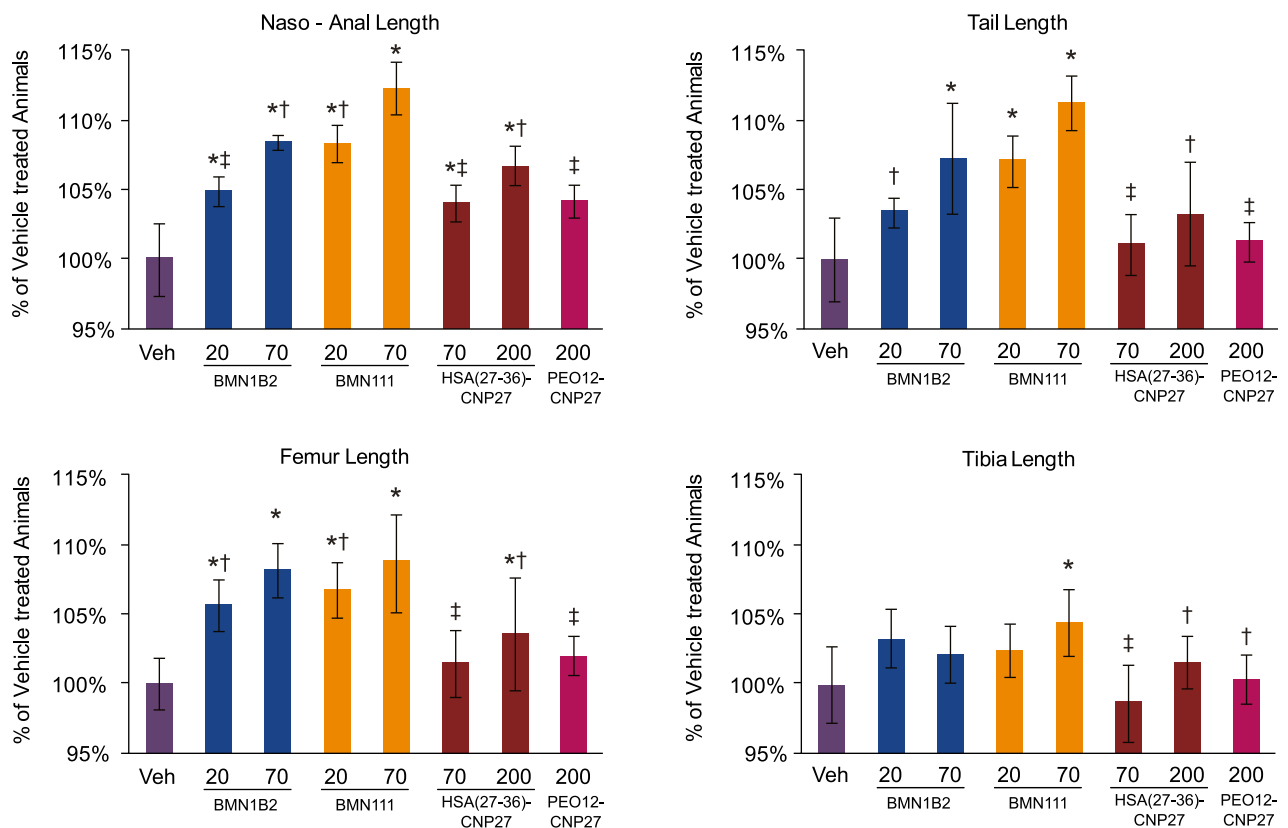


Fig. 3. Wild-type (FVB/nJ) mice treated with various NEP-resistant CNP variants. Growth of the appendicular and axial skeletons of wild-type mice (FVB/nJ) treated with CNP variants. Three-week-old wild-type mice were given daily subcutaneous administrations of CNP variants (20, 70 or 200 nmol/kg; $n = 8$ /group) or vehicle for 5 weeks. The asterisk denotes statistical significance compared with the vehicle control ($P < 0.05$; ANOVA with post hoc Dunnett's t test). The dagger denotes significance compared with BMN 111 at 70 nmol/kg. The double dagger denotes significance compared with BMN 111 at both 20 and 70 nmol/kg (one-way ANOVA, post hoc Tukey's test). ANOVA, analysis of variance; Veh, vehicle. Error bars indicate the S.D.

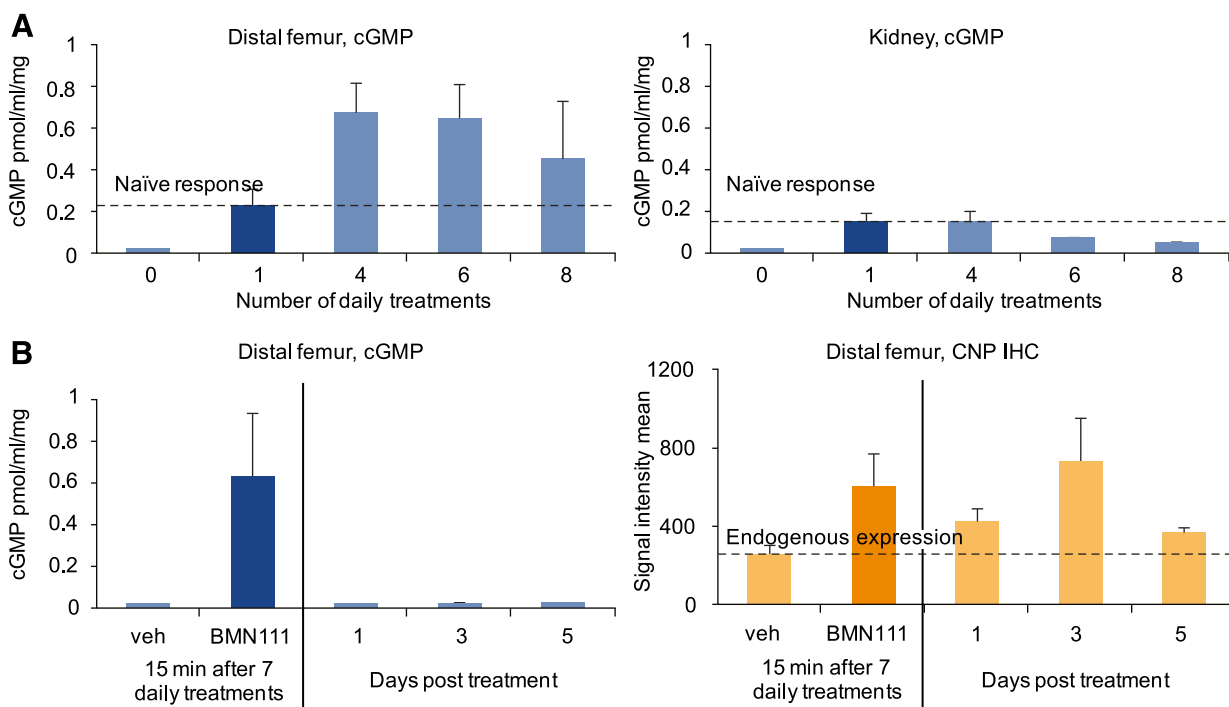


Fig. 4. Activity, accumulation, and clearance of NEP-resistant CNP variant, BMN 111, at the growth plate. (A) cGMP production during daily treatments. Wild-type CD1 mice were treated with 200 nmol/kg BMN 111 daily for as long as 8 days. Distal femura, containing the growth plate, and kidneys were dissected 15 minutes after the first, fourth, sixth, and eighth doses and cGMP was extracted and quantified ($n = 2$). (B) BMN 111 residence and activity during after treatment withdrawal. Wild-type CD1 mice were treated daily with 200 nmol/kg BMN 111 for 7 days. Samples were obtained after treatment withdrawal. Distal femora, containing the growth plate, were dissected 15 minutes after the last treatment and 1, 3, and 5 days thereafter ($n = 2$). Tissues were used for cGMP analysis or CNP IHC. Confocal microscopy allowed for detection of accumulated CNP signal in defined regions of interest of the growth plate. cGMP was quantified by a competitive enzyme-linked immunosorbent assay and normalized for tissue weight. veh, vehicle. Error bars indicate the S.D.

Pharmacological Effects of BMN 111 in ACH Mice.

Targeted expression of an activated FGFR3 in the growth plate cartilage of mice was achieved using regulatory elements of the collagen 2 gene (Naski et al., 1998). Three-week-old

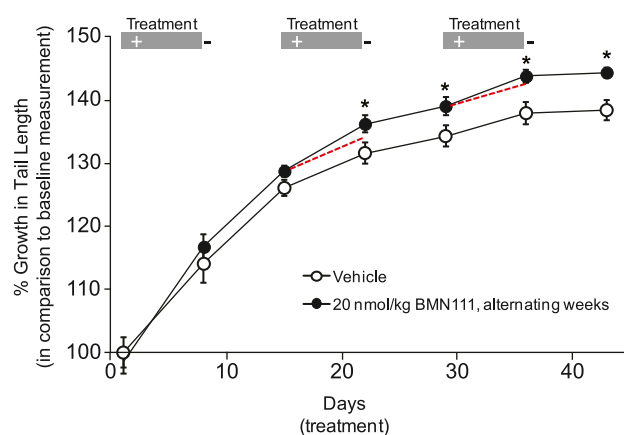


Fig. 5. Effect of discontinuous BMN 111 dose intervals on axial skeletal growth. Three-week-old wild-type (FVB/nJ) male mice were given subcutaneous injections of BMN 111 (20 nmol/kg) daily on alternating weeks (weeks 1, 3, and 5) or vehicle daily for 5 weeks. Tail measurements were collected at study initiation. A normal growth pattern resumes after discontinuation of treatment. Statistical significance (* $P < 0.05$ versus vehicle) was noted for all end points beginning at day 22 through the end of the study (analysis of variance with post hoc Dunnett's test). The red dotted lines depict normal growth and were added to illustrate accelerated growth during the treatment period ($n = 10$ /group). Error bars indicate the S.D.

Fgfr3^{ACH/+} male mice and their wild-type (FVB/nJ) littermates ($n = 8$ –10/group) were given daily subcutaneous injections of BMN 111 at 5, 20, or 70 nmol/kg (20, 80, or 280 μ g/kg) or vehicle for 36 days. Significant growth in the appendicular and axial skeletons was observed in BMN 111-treated *Fgfr3*^{ACH/+} mice (Fig. 7A). Although this *Fgfr3*^{ACH/+} mouse model represents a mild phenotype, naso-anal and femur lengths of *Fgfr3*^{ACH/+} mice were significantly shorter than wild-type mice at the start of the study ($P < 0.05$). Correction of the tail length was observed after 36 days of BMN 111 daily subcutaneous administrations at the 70 nmol/kg dose level. Naso-anal lengths were corrected at 20 nmol/kg after daily subcutaneous administration of BMN 111 for 36 days. Femur and tibia lengths were corrected at 5 and 20 nmol/kg by the end of the study. Histologic examination revealed a statistically significant increase in growth plate expansion in *Fgfr3*^{ACH/+} mice treated with 70 nmol/kg BMN 111 (Fig. 7B), including increased area and/or height in the zones of resting cartilage, proliferation, and hypertrophy (data not shown). These data indicate that BMN 111 activation of NPR B corrects growth plate abnormalities secondary to the *Fgfr3* mutation that results in ACH dwarfism.

Hemodynamic Effects of BMN 111 in Cynomolgus Monkeys. After initial dose-ranging studies were performed in mice (Fig. 6), a pilot study was performed in normal juvenile, anesthetized or conscious, cynomolgus monkeys after a single subcutaneous administration (dose range, 0–35 nmol/kg), measuring acute cardiovascular effects of BMN 111, to determine the doses to be used in a long-term (6-month) study looking at growth and tolerability parameters (Fig. 8). The aim

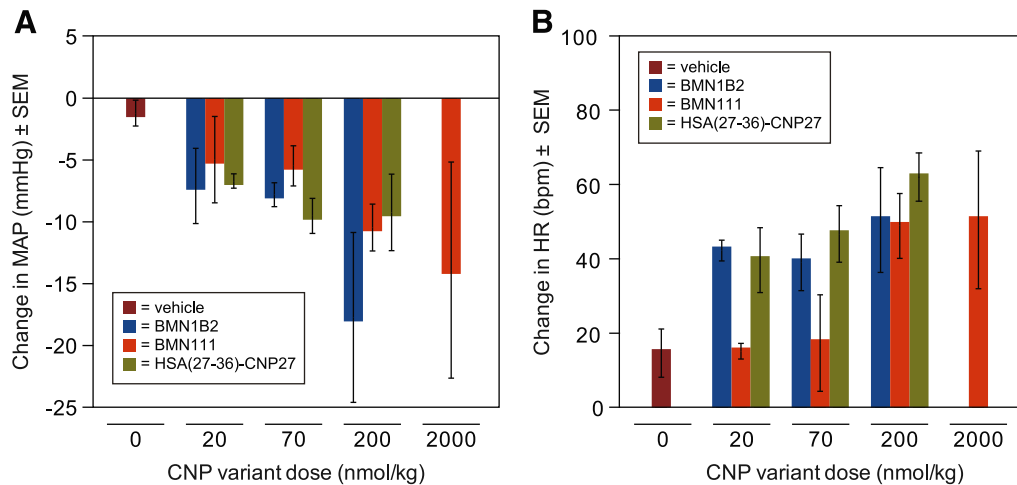


Fig. 6. (A and B) Change in MAP (A) and HR (B) in anesthetized mice treated with NEP-resistant CNP variants. CNP variants were tested over a dose range of 20–200 nmol/kg (2000 nmol/kg for BMN 111) in 6- to 7-week-old wild-type mice ($n = 3/\text{group}$; vehicle $n = 5/\text{group}$). The difference between mean values over the 15 minutes predose and 15 minutes postdose is shown; this encompassed the time of BP nadir and HR zenith. bpm, beats per minute.

was to find a tolerable dose that yielded a ≤ 10 -mm Hg (approximately 10%) decrease in MAP or a ≤ 50 -beats per minute increase (approximately 25%) in resting HR. It was observed that HR increase was the most sensitive parameter, probably due to reflex tachycardia, and a dose of 7 nmol/kg gave an approximately 25% increase in HR in conscious monkeys, with little or no effect on MAP (Fig. 8, B and D). The increase in HR was transient, with the maximal increase observed at 10–20 minutes postdose (Fig. 8, E and F). Multiple subcutaneous daily dosages (7 and 17.5 nmol/kg) for 7 consecutive days were well tolerated. ECG parameters were unaffected at any dose of BMN 111 tested (data not shown). The drop in MAP after BMN 111 administration was inconsistent and often somewhat less marked after subsequent doses (data not shown). On the basis of these data, the highest dose chosen for the long-term study was 8.25 nmol/kg. A lower dosage of 2.25 nmol/kg per day, which gave little or no cardiovascular effect, was also tested.

Pharmacological Effects of BMN 111 in Cynomolgus Monkeys. BMN 111 was administered subcutaneously to growing (2- to 4-year-old) cynomolgus male monkeys at 2.25 or 8.25 nmol/kg once daily for 6 months ($n = 4/\text{group}$). Although BP was not monitored, no clinical signs of hypotension or distress were noted in any animal at any time during the study. The effect on proximal tibial growth plate size was observed by MRI imaging performed during the fourth week of dosing (Fig. 9A). Mean growth plate volume increased approximately 40% for the high-dose group versus the pretreatment volume. This was the peak growth plate volume noted. Volume receded thereafter, but remained greater than baseline throughout the remainder of the 6-month study. Treatment with BMN 111 resulted in a dose-dependent increase in total tibial length (measured from digital radiographs) and rate of growth (Fig. 9b) as well as increased lengths of arms, legs, and tail when measured externally (data not shown). Treated animals maintained their height/length advantage through the end of the study period. Clinical chemistry and hematology parameters remained normal and unchanged throughout the 6-month study with the noted exception of increased serum levels of total and bone-specific alkaline phosphatase associated with the increase in bone formation (Fig. 9C).

Growth plate expansion, evaluated post mortem after 6 months of treatment, was evident at the histologic level (Fig. 10B, upper), with significant expansions in total growth plate thickness, proliferating zone thickness, and hypertrophic zone thickness, changes that are associated with inhibition of FGFR3 signaling (Iwata et al., 2000; Ornitz and Marie, 2002) (Table 3). Similar histologic and growth plate expansion results were observed in wild-type and *Fgfr3^{ACH/+}* murine studies (Fig. 10A; Table 3). Double fluorochrome labeling of newly formed mineralized bone performed during the final 14 days of the in vivo study illustrated that growth plate expansion in response to 8.25 nmol/kg per day BMN 111 translated into increased longitudinal growth of mineralized bone (Fig. 10B, lower panel). Static and dynamic measurements of trabecular bone architecture and turnover were not affected by BMN 111 treatment, indicating that normal bone was formed (Table 4).

To assess the effects of BMN 111 treatment on vertebral foramen area, post mortem microcomputed tomography was performed on excised lumbar vertebrae 2–4. For the high-dose group (8.25 nmol/kg per day), mean vertebral foramen area increased approximately 10%–17% going up the spine (L4 to L2) versus the vehicle control group, and was statistically significant in L2 ($P = 0.03$ versus vehicle by two-tailed t test) (Fig. 9D).

Discussion

In ACH, mutations in FGFR3 result in constitutive activation, suppressing the proliferation and differentiation of chondrocytes resulting in improper cartilage to bone conversion in the growth plate (Laederich and Horton, 2010). ACH is associated with significant morbidity and increased mortality, and current treatments are mostly surgical (Trotter and Hall, 2005; Wynn et al., 2007). BMN 111, a CNP variant, offers a potential treatment for ACH that addresses the underlying biochemical defect. By signaling through NPR B, BMN 111 suppresses downstream signals in normal and mutated FGFR3 pathways to enhance or restore chondrocyte proliferation and differentiation resulting in bone growth. Specifically, BMN 111 inhibits the ERK/MAPK pathway through phosphorylation of Raf-1 by cGMP-dependent protein kinase 2 (Krejci et al., 2005).

A

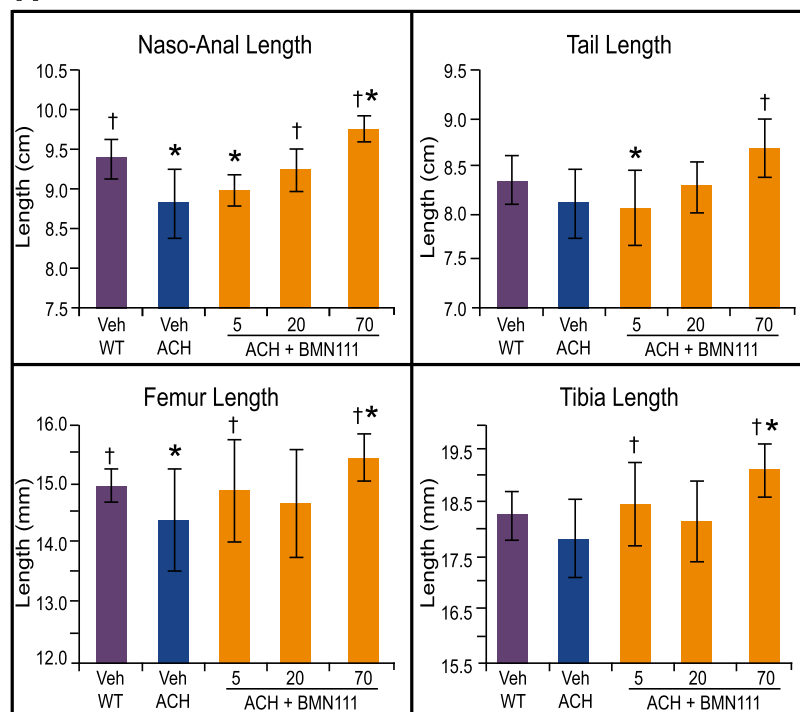
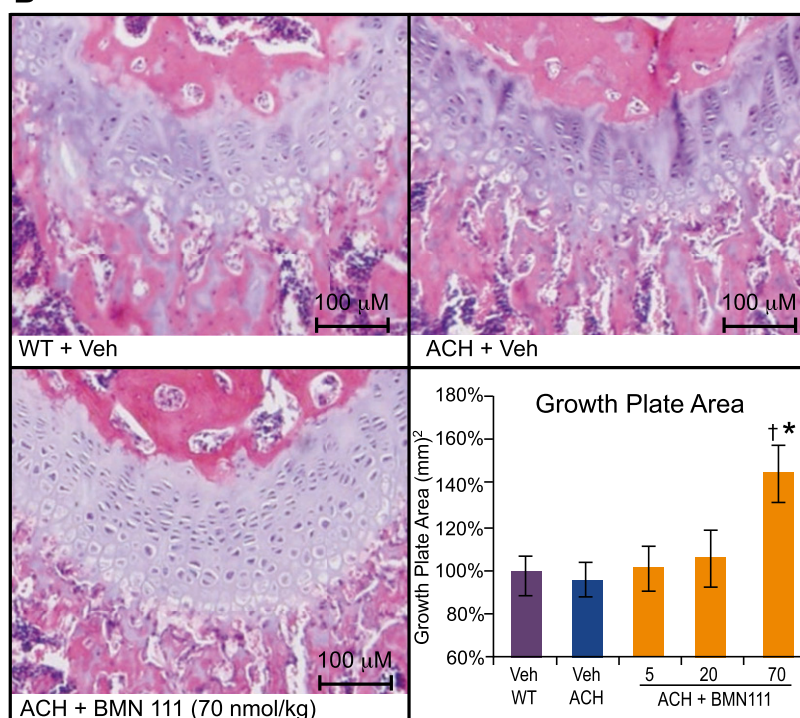


Fig. 7. *Fgfr3^{ACH/+}* mice treated with BMN 111. (A) Growth of the appendicular and axial skeletons of *Fgfr3^{ACH/+}* mice after treatment with BMN 111. Three-week-old *Fgfr3^{ACH/+}* mice given daily subcutaneous administrations of BMN 111 (5, 20, or 70 nmol/kg) or vehicle for 5 weeks ($n = 8$ /group). Wild-type vehicle controlled mice (FVB/nJ) were included to assess the degree of phenotype and normalization for each growth parameter ($n = 8$). The asterisk denotes statistical significance ($P < 0.05$) against vehicle-treated wild-type mice. The dagger denotes statistical significance against vehicle-treated *Fgfr3^{ACH/+}* mice (analysis of variance with post hoc Dunnett's t test). (B) Distal femoral growth plates of mice treated with vehicle or BMN 111 (trichrome stained). Significant growth plate expansion was observed in *Fgfr3^{ACH/+}* mice treated with 70 nmol/kg BMN 111. Error bars indicate the S.D. Original magnification, 10 $\times$ in B. Veh, vehicle; WT, wild type.

B



Because CNP is rapidly cleared from the circulation through receptor-mediated (NPR C) and proteolytic (NEP) pathways (Potter, 2011), CNP requires continuous infusion to be effective in ACH murine studies (Yasoda et al., 2009); however, this is not a desired therapy by physicians or patients. To overcome these limitations, we developed a CNP variant, BMN 111, which resists degradation by NEP at the site of subcutaneous administration and at the growth plate (Ruchon et al., 2000;

Yamashita et al., 2000; Nakajima et al., 2012). Here, we demonstrate that BMN 111 is effective as a subcutaneous injectable therapeutic that promotes bone growth in juvenile wild-type mice and juvenile cynomolgus monkeys and corrects the ACH phenotype in *Fgfr3^{ACH/+}* mice.

NEP prefers small peptides based on physiologic substrate and crystal structure data. Larger CNP variants (>3 kDa) demonstrated in vitro NEP resistance and a subset retained

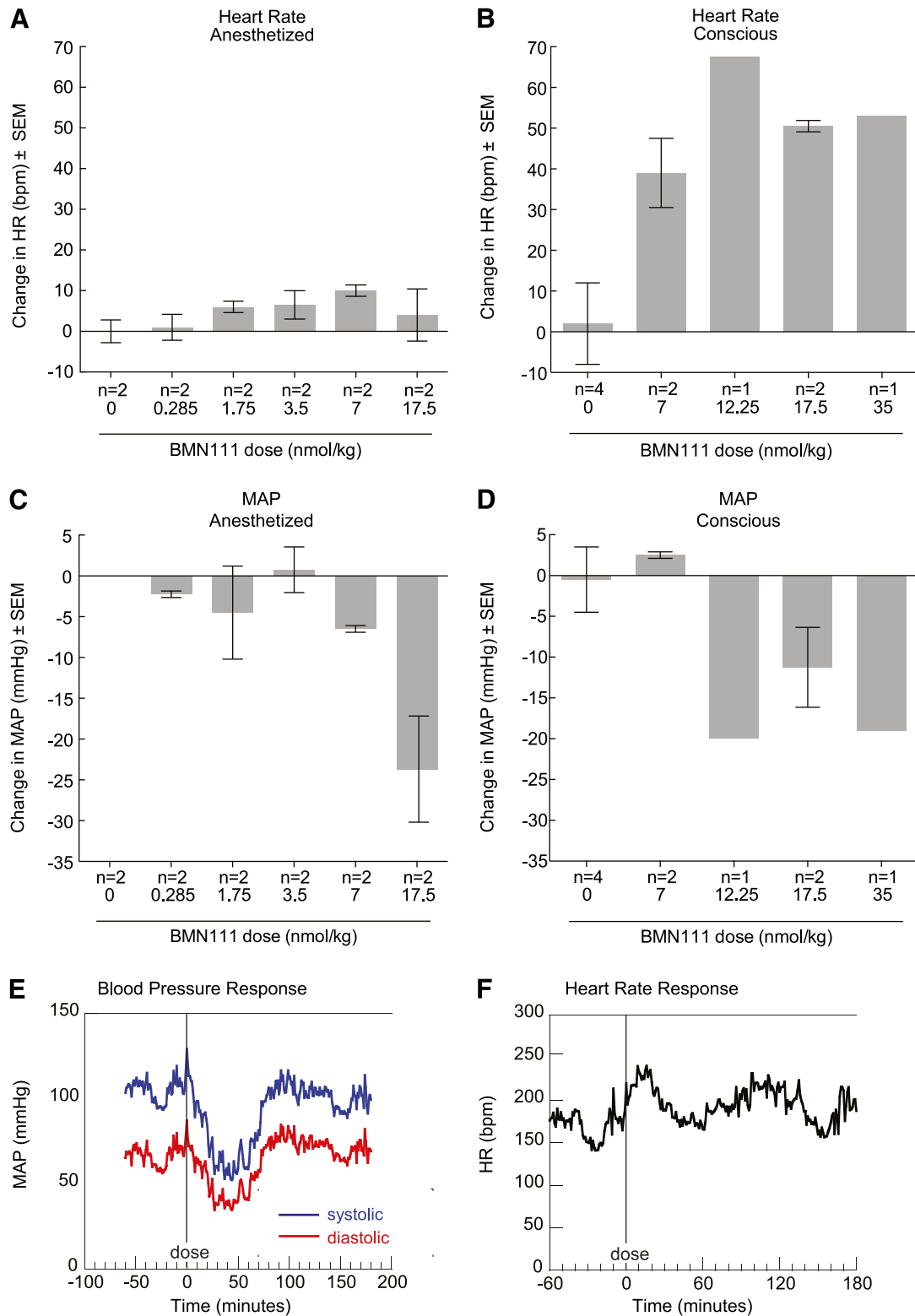


Fig. 8. Effect of BMN 111 on BP and HR in cynomolgus monkeys. (A–D) In both anesthetized (A and C) and conscious monkeys (B and D), BMN 111 decreased MAP in a dose-dependent manner ($n = 1\text{--}4/\text{group}$). In conscious animals, there was a concomitant increase in HR. The HR response was blunted in the anesthetized animals. (A and B) Change in average HR over 10–20 minutes postdose (encompassing time of HR zenith) and baseline (15 minutes just prior to dosing). (C and D) Change in average MAP over 10–20 minutes postdose (encompassing time of MAP nadir) and baseline average (15 minutes just prior to dosing). (E and F) BP (E) and HR (F) after a single subcutaneous dose of BMN111 (17.5 nmol/kg) to a conscious monkey. Significant hypotension develops rapidly after administration, but begins to resolve within an hour. bpm, beats per minute.

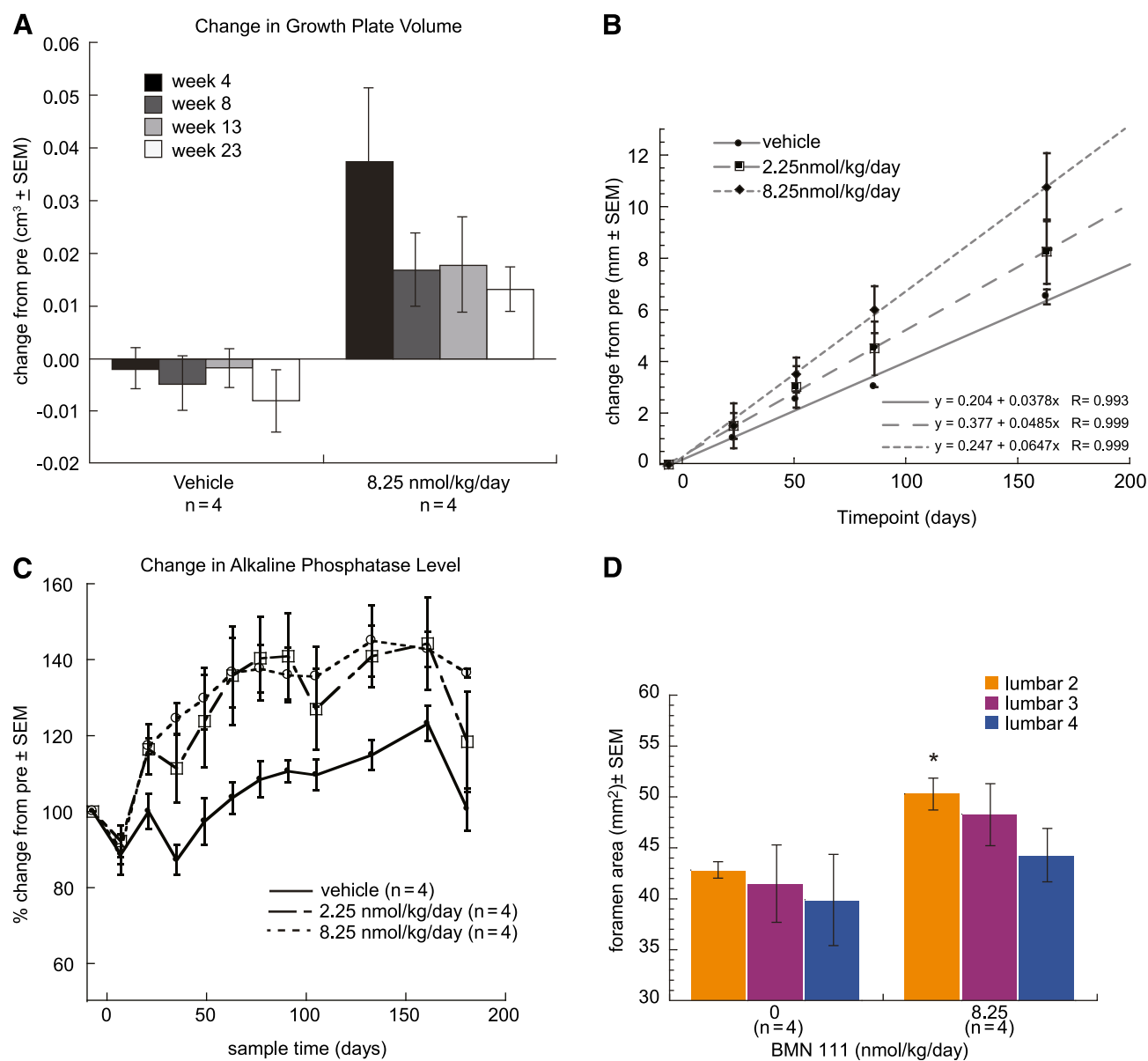


Fig. 9. Change in growth plate volume, tibial length, serum alkaline phosphatase levels, and lumbar vertebral foramen in cynomolgus monkeys treated with BMN 111. (A) Change in right proximal tibial growth plate volume with high-dose BMN 111 treatment, measured by MRI ($P = \text{N.S.}$ versus vehicle at all time points; $n = 4/\text{group}$). (B) Radiographic evaluation of cynomolgus tibias at several time points in animals treated with BMN 111. Dose-dependent change in rate of growth of tibial length. Right tibial lengths (in millimeters) were measured manually on posterior–anterior projections with dedicated image analysis software ($P = \text{N.S.}$ versus vehicle at all time points ($n = 4/\text{group}$)). (C) Increase in serum alkaline phosphatase with BMN 111 treatment. Known as markers of bone growth or deposition, changes in both total and bone-specific alkaline phosphatase (data not shown) were not statistically significant over prestudy values ($n = 4/\text{group}$). (D) Area of lumbar vertebral foramen of cynomolgus monkey assessed by microcomputed tomography. In vertebrae L2, L3, and L4, treatment with BMN111 at the high dose resulted in a trend toward greater area of vertebral foramen compared with vehicle controls. For L2, the increase was statistically significant ($*P = 0.03$ versus vehicle; $n = 4/\text{group}$). N.S., not significant.

native CNP in vitro activity. NEP resistance translated into improved serum half-lives in wild-type rat or murine studies ($t_{1/2} =$ approximately 15 minutes for CNP variants versus 2–5 minutes for native CNP); however, the improved in vivo stability does not exclude the possibility that CNP variants are susceptible to other proteolytic pathways in addition to the known NPR clearance pathway (NPR C) present in the vasculature. BMN 111 was selected for ACH murine studies and larger animal studies based on its superior bone growth-promoting attributes in the wild-type murine studies. The lowest dosage tested in the wild-type murine screening study was 20 nmol/kg per day and this appeared to be well above the

minimal effective dose. This dose also appeared to correct most growth deficits in the *Fgfr3*^{ACH/+} mouse model. Importantly, we recently reported that BMN 111 stimulated bone growth in mouse models containing a stronger activating mutation of *Fgfr3* (*Fgfr3*^{Y367C/+}), a mutation that results in thanatophoric dysplasia type I (TDI) in humans (Lorget et al., 2012).

To test its effectiveness in larger animals, levels that had minimal effects on hemodynamic parameters were chosen and three cohorts of cynomolgus monkeys were dosed. Dose-dependent growth was observed in this 6-month study. The high-dose group showed measurable increases in growth plate expansion, rate of endochondral bone growth, and trends in

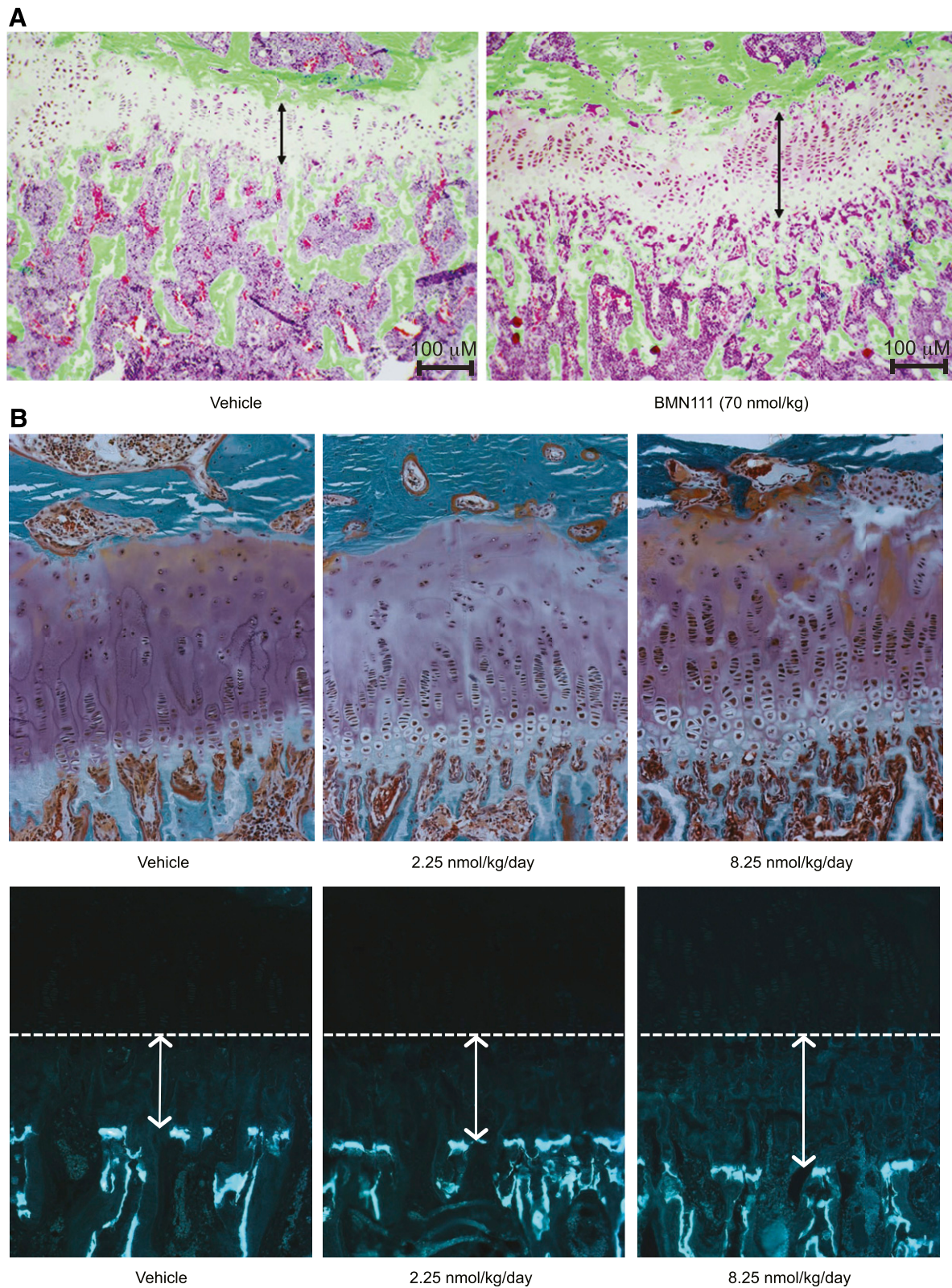


Fig. 10. Cynomolgus monkey and wild-type mice growth plate histology after 6 months of treatment with BMN 111. (A) Distal femoral growth plates of mice treated with vehicle or BMN 111 (trichrome stained). Growth plate expansion was observed in mice treated with 20 and 70 nmol/kg BMN 111 (showing 70 nmol/kg). (B) The upper panel shows Goldner trichrome staining of growth plate (purple) and bone (green). The lower panel is the calcein label under UV (green) showing longitudinal growth rate in the last 14 days of treatment. The distal edge of growth plate is delineated with a dashed line, and longitudinal bone growth in 14 days prior to necropsy is represented with arrows ($n = 4/\text{group}$, showing one representative image from each group). Original magnification, $10\times$ in A.

TABLE 3
Growth plate parameters and longitudinal growth rates
Data are presented as the mean $\pm$ SD, with *P* values in parentheses. *P* values for cynomolgus monkeys were obtained using analysis of variance with Tukey post hoc analysis versus vehicle, and *P* values for wild-type mice and *Fgfr3*^{Ach/+} mice were obtained using analysis of variance with post hoc Dunnett's test.

Parameter	Cynomolgus Monkey			Wild-Type Mice			<i>Fgfr3</i> ^{Ach/+} Mice		
	Vehicle (<i>n</i> = 4)	2.25 nmol/kg per day (<i>n</i> = 4)	8.25 nmol/kg per day (<i>n</i> = 4)	Vehicle (<i>n</i> = 10)	70 nmol/kg per day (<i>n</i> = 10)	Vehicle (<i>n</i> = 8)	5 nmol/kg per day (<i>n</i> = 5)	20 nmol/kg per day (<i>n</i> = 5)	70 nmol/kg per day (<i>n</i> = 4)
Longitudinal growth rate (μ m/day)	26 $\pm$ 7	26 $\pm$ 5 (N.S.)	40 $\pm$ 9 (<0.05)						
Growth plate thickness (μ m) <i>Fgfr3</i> ^{Ach/+} mice	555 $\pm$ 61	594 $\pm$ 64 (N.S.)	682 $\pm$ 48 (<0.05)	159.4 $\pm$ 17.0	200.7 $\pm$ 14.4 (<0.001)		134.0 $\pm$ 22.5 (N.S.)	142.2 $\pm$ 20.2 (N.S.)	163.2 $\pm$ 24
Wild-type mice						137.3 $\pm$ 12.7			(N.S.)
Proliferating zone thickness (μ m)	125 $\pm$ 10	139 $\pm$ 89 (N.S.)	196 $\pm$ 14 (<0.001)			125.2 $\pm$ 15.1			(<0.05)
Proliferating cells/ column (<i>n</i>)	13 $\pm$ 2	11 $\pm$ 2 (N.S.)	11 $\pm$ 1.6 (N.S.)						
Hypertrophic zone thickness (μ m)	72 $\pm$ 26	89 $\pm$ 23 (N.S.)	128 $\pm$ 56 (<0.05)						
Hypertrophic cell volume (μ m ²)	232 $\pm$ 30	258 $\pm$ 56 (N.S.)	286 $\pm$ 34 (N.S.)						

expansion of the vertebral foramen. Although this study was not powered for significance, some statistically significant trends were observed; for example, growth plate thickness in the high-dose group, particularly in the proliferating and hypertrophic zones, was statistically larger than in vehicle-treated animals ($P < 0.001$ and $P < 0.05$, respectively), which is consistent with observations that suggest that FGFR3 inhibits both the proliferation and terminal differentiation of growth plate chondrocytes and the synthesis of extracellular matrix by these cells (Laederich and Horton, 2010). Double fluorochrome labeling of newly formed mineralized bone demonstrated that bone formation was increased in the high-dose group in accordance with the increased endochondral activity that caused growth plate expansion. Moreover, the achievement of this bone growth in the last 14 days of the study demonstrated the continued effectiveness of BMN 111 after chronic treatment, and suggested that the growth plate width, which receded after 4 weeks of treatment, was not associated with a reduction of BMN 111 activity. BMN 111-treated animals showed equivalent trabecular architecture parameters compared with vehicle-treated animals, suggesting that BMN 111 treatment did not significantly affect osteoclast activity, if at all.

Several other groups have reported potential therapeutic strategies that modulate the aberrant FGFR3 pathway. Garcia et al., 2013 demonstrated that a soluble fibroblast growth factor receptor-3 (sFGFR3) could act as a decoy receptor to prevent fibroblast growth factor (FGF) from binding to and signaling through the FGFR3. In vitro binding studies with fixed concentrations of FGF2, FGF9, and FGF18 demonstrated that sFGFR3 was required in 100-fold excess to reduce the concentration of these FGFs by one-half. Nevertheless, they were able to show stimulation of bone growth in wild-type and *Fgfr3*^{Ach/+} murine studies. The long-term effects of continuous FGF depletion remain to be determined, but would be expected to impair wound repair and other developmental processes (Lynch et al., 1989; Kurtz et al., 2004). One question that comes to mind with this therapeutic strategy is whether sufficient amounts of this approximately 70-kDa sFGFR3 protein could diffuse through the highly negatively charged extracellular matrix of a larger human growth plate to compete for FGFs expressed locally as paracrine factors. Moreover, there is still no scientific consensus that FGF receptors require ligand for dimerization (He et al., 2011; Placone and Hristova, 2012).

In another report, Jin et al. (2012) discovered a 12-amino acid peptide through phage display, P3, which could bind to the extracellular domain of FGFR3 and partially block FGF2-mediated ERK1/2 phosphorylation. When pregnant *Fgfr3*^{Neo-K644E/+} mice [phenotypically normal thanatophoric dysplasia type II (TDII) carriers] were given daily peritoneal injections of P3 (100 μ g/kg body weight) at E16.5 until birth, all TDII pups (*Fgfr3*^{K644E/+}) survived, whereas all vehicle control TDII pups died. The TDII mice that survived had increased thoracic cavities, which rescued the postnatal lethality phenotype; however, the rescued mice still had smaller bodies and dome-shaped skulls compared with their wild-type littermates. P3 as a postnatal therapy for ACH, perhaps a more acceptable therapeutic regimen, was not tested in this study.

Matsushita et al. (2013) identified meclozine, an anti-histamine used for motion sickness, as an antagonist of the

TABLE 4

Trabecular architecture parameters: histomorphometric analysis of the left proximal tibial trabecular bone of cynomolgus monkeys treated with BMN 111 or vehicle
Data are presented as the mean $\pm$ S.D. ($n = 4$). There were no significant differences (analysis of variance) between groups for all parameters.

Parameter	Vehicle	2.25 nmol/kg per day	8.25 nmol/kg per day
Bone volume/tissue volume (%)	22 $\pm$ 5	27 $\pm$ 7	29 $\pm$ 6
Osteoid/bone surface (%)	33 $\pm$ 9	33 $\pm$ 6	33 $\pm$ 9
Trabecular thickness (μm)	133 $\pm$ 17	158 $\pm$ 24	132 $\pm$ 11
Trabecular number (mm^{-1})	1.6 $\pm$ 0.4	1.7 $\pm$ 0.3	2.2 $\pm$ 0.3
Trabecular spacing (μm)	501 $\pm$ 137	452 $\pm$ 124	339 $\pm$ 83
Osteoblasts/bone surface (n)	22 $\pm$ 2.4	23 $\pm$ 4	25 $\pm$ 3.8
Osteoclasts/bone surface (n)	1.8 $\pm$ 0.5	1.4 $\pm$ 0.5	1.3 $\pm$ 0.7
Osteoid thickness (μm)	8.2 $\pm$ 2.1	9 $\pm$ 1.3	9.4 $\pm$ 1.4
Mineral apposition rate per day ($\mu\text{m}/\text{day}$)	1.6 $\pm$ 0.1	1.9 $\pm$ 0.3	1.7 $\pm$ 0.4
Bone formation rate/bone volume	0.013 $\pm$ 0.002	0.015 $\pm$ 0.003	0.011 $\pm$ 0.002

FGFR3 pathway. They demonstrated that meclozine was able to attenuate FGF2-mediated ERK phosphorylation in rat chondrosarcoma cells, facilitate chondrocytic differentiation of ATDC5 cells expressing ACH or TDII mutant FGFR3, and promote tibial growth in FGF2-suppressed tissue explant studies. In explant studies, they compared CNP (0.2 μM) to meclozine (20 μM). Interestingly, meclozine demonstrated no statistically significant enhancement of tibial growth in the absence of FGF2, unlike CNP (Yasoda et al., 1998). Furthermore, meclozine was not tested in any of the available in vivo murine models for its ability to stimulate or correct growth. Thus, questions remain as to whether this is a viable therapeutic option.

In a recent article, Yamashita et al. (2014) demonstrated that statins could correct the degraded cartilage in both chondrogenically differentiated TDI and ACH induced pluripotent stem cells. Interestingly, mRNA expression levels of FGFR3 were increased by lovastatin, but protein levels by immunoblot decreased, which led the authors to postulate that statins increase the degradation rate of FGFR3 in chondrogenically differentiated TDI induced pluripotent stem cells. In an 11-day ACH murine study (days 3–14), mice receiving daily injections of rosuvastatin demonstrated an increase in distal long-bone growth rate comparable to wild-type mice receiving vehicle. The effect beyond 14 days on final growth (6–8 weeks) was not assessed in this study. The mechanism is unclear but could be due to altering membrane dynamics, which may not be a good strategy given the frequency of known side effects of statins as well as the potential developmental consequences (Evans and Rees, 2002; Maji et al., 2013).

We believe that BMN 111 is a promising therapeutic option for children with ACH with open growth plates for a number of reasons. First, BMN 111, an NEP-resistant CNP variant, is a natural antagonist of the FGFR3 pathway, corrects the phenotype in *Fgfr3*^{ACH/+} mice, and attenuates the phenotype in stronger activating mutations of FGFR3 (TDI; Y367C/+) when given daily subcutaneously (Lorget et al., 2012). Second, CNP and its receptor are expressed in the growth plate. Third, the amino acid content is basic (pI = approximately 10) and the peptide is small, which enable subcutaneous administered BMN 111 to target and diffuse through the anionic extracellular matrix barrier of the growth plate. Finally, unlike other small molecule strategies, BMN 111 will only target cells that express its cognate receptor, NPR B, which should mitigate many of the side effects seen with these other approaches. It should be noted here that NPR B is not limited to the growth plate, but humans lacking NPR B have a dwarfism without any other apparent disease (Bartels et al., 2004). Overactive NPR B produces tall

stature, scoliosis, and great toe macrodactyly, but apparently nothing else (Toydemir et al., 2006).

An additional unique feature of CNP is that it increases proteoglycan synthesis independent of the FGF receptor/ERK pathway (Krejci et al., 2005; Waldman et al., 2008), which may be partly responsible for the anabolic bone growth effects observed in wild-type mice and normal cynomolgus monkeys. Recent evidence suggests that agonists of the decoy receptor NPR C, such as CNP, may also be contributing to these anabolic effects (Peake et al., 2013). On the basis of these findings and our data in wild-type mice and normal cynomolgus monkeys, it is conceivable that BMN 111 could be used to treat other FGFR3-related skeletal dysplasias, such as hypochondroplasia, and perhaps idiopathic short stature, in which no clear causal mechanism has been ascribed. BMN 111 is currently being investigated in children with ACH (ClinicalTrials.gov identifier NCT02055157).

In conclusion, through a series of in vitro and in vivo rodent studies, we identified five CNP variants comprising three types (PEGylated, chimeric, and natural amino acid extensions) that were resistant to NEP by virtue of size, retained native in vitro potency, and demonstrated prolonged half-lives in rats and mice. One CNP variant, BMN 111, was selected for further study based on potency and similarity to native CNP. When administered subcutaneously to normal mice, normal growing monkeys, or ACH mice, BMN 111 treatment resulted in growth of the axial and appendicular skeletons. In the 6-month daily dose study in juvenile monkeys, BMN 111 (administered at doses that did not cause an unacceptable hemodynamic effect) resulted in expansion of the proximal tibial growth plates, with widening of the hypertrophic zone, increased length and rate of limb growth, and increased area of the foramen of lumbar vertebrae. Concomitant increase in both total and bone-specific alkaline phosphatase levels may provide a biomarker of early BMN 111 activity. Transient, asymptomatic dose-dependent hemodynamic responses were observed in mice and monkeys at doses higher than needed to produce skeletal growth. These experiments indicate that growth in both normal and ACH juvenile animals is governed, at least in part, through the NPR B cGMP signaling pathway, and that BMN 111 affects this pathway. BMN 111 is being investigated as a potential therapeutic for pediatric patients with ACH.

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Conducted experiments: Wendt, Dvorak-Ewell, Bullens, Lorget, Bell, Castillo, Aoyagi-Scharber, Krejci, Bunting.

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Wrote or contributed to the writing of the manuscript: Wendt, Dvorak-Ewell, Bullens, Bell.

References

- Abbey SE and Potter LR (2003) Lysophosphatidic acid inhibits C-type natriuretic peptide activation of guanylyl cyclase-B. *Endocrinology* **144**:240–246.
- Barr CS, Rhodes P, and Struthers AD (1996) C-type natriuretic peptide. *Peptides* **17**:1243–1251.
- Bartels CF, Bükümmez H, Padayatti P, Rhee DK, van Ravenswaaij-Arts C, Pauli RM, Mundlos S, Chitayat D, Shih LY, Al-Gazali LI, et al. (2004) Mutations in the transmembrane natriuretic peptide receptor NPR-B impair skeletal growth and cause acromesomelic dysplasia, type Maroteaux. *Am J Hum Genet* **75**:27–34.
- Boccardi R, Giorda R, Buttgeriet J, Gimelli S, Divizia MT, Beri S, Garofalo S, Tavella S, Lerone M, Zuffardi O, et al. (2007) Overexpression of the C-type natriuretic peptide (CNP) is associated with overgrowth and bone anomalies in an individual with balanced t(2;7) translocation. *Hum Mutat* **28**:724–731.
- Brandt RR, Heublein DM, Aarhus LL, Lewicki JA, and Burnett JC (1995) Role of natriuretic peptide clearance receptor in vivo control of C-type natriuretic peptide. *Am J Physiol* **269**:H326–H331.
- Brandt RR, Mattingly MT, Clavell AL, Barclay PL, and Burnett JC, Jr (1997) Neutral endopeptidase regulates C-type natriuretic peptide metabolism but does not potentiate its bioactivity in vivo. *Hypertension* **30**:184–190.
- Charles CJ, Espiner EA, Richards AM, Nicholls MG, and Yandle TG (1995) Biological actions and pharmacokinetics of C-type natriuretic peptide in conscious sheep. *Am J Physiol* **268**:R201–R207.
- Chusho H, Tamura N, Ogawa Y, Yasoda A, Suda M, Miyazawa T, Nakamura K, Nakao K, Kurihara T, Komatsu Y, et al. (2001) Dwarfism and early death in mice lacking C-type natriuretic peptide. *Proc Natl Acad Sci USA* **98**:4016–4021.
- Clavell AL, Stingo AJ, Wei CM, Heublein DM, and Burnett JC, Jr (1993) C-type natriuretic peptide: a selective cardiovascular peptide. *Am J Physiol* **264**:R290–R295.
- Dunnett CW and Crisafio R (1955) The operating characteristics of some official weight variation tests for tablets. *J Pharm Pharmacol* **7**:314–327.
- Erdős EG and Skidgel RA (1989) Neutral endopeptidase 24.11 (enkephalinase) and related regulators of peptide hormones. *FASEB J* **3**:145–151.
- Evans M and Rees A (2002) The myotoxicity of statins. *Curr Opin Lipidol* **13**:415–420.
- Foldynova-Trantirkova S, Wilcox WR, and Krejci P (2012) Sixteen years and counting: the current understanding of fibroblast growth factor receptor 3 (FGFR3) signaling in skeletal dysplasias. *Hum Mutat* **33**:29–41.
- Furuya M, Tawaragi Y, Minamitake Y, Kitajima Y, Fuchimura K, Tanaka S, Minamino N, Kangawa K, and Matsuo H (1992) Structural requirements of C-type natriuretic peptide for elevation of cyclic GMP in cultured vascular smooth muscle cells. *Biochem Biophys Res Commun* **183**:964–969.
- Garcia S, Dirat B, Tognacci T, Rochet N, Mouska X, Bonnafous S, Patoureaux S, Tran A, Gual P, Le Marchand-Brustel Y, et al. (2013) Postnatal soluble FGFR3 therapy rescues achondroplasia symptoms and restores bone growth in mice. *Sci Transl Med* **5**:203ra124.
- He L, Shobnam N, Wimley WC, and Hristova K (2011) FGFR3 heterodimerization in achondroplasia, the most common form of human dwarfism. *J Biol Chem* **286**:13272–13281.
- Hecht JT and Butler LJ (1990) Neurologic morbidity associated with achondroplasia. *J Child Neurol* **5**:84–97.
- Horton WA, Hall JG, and Hecht JT (2007) Achondroplasia. *Lancet* **370**:162–172.
- Hunt PJ, Richards AM, Espiner EA, Nicholls MG, and Yandle TG (1994) Bioactivity and metabolism of C-type natriuretic peptide in normal man. *J Clin Endocrinol Metab* **78**:1428–1435.
- Ikagi T, Itoh H, Suga SI, Hama N, Ogawa Y, Komatsu Y, Yamashita J, Doi K, Chun TH, and Nakao K (1998) Effects of intravenously administered C-type natriuretic peptide in humans: comparison with atrial natriuretic peptide. *Hypertens Res* **21**:7–13.
- Iwata T, Chen L, Li C, Ovchinnikov DA, Behringer RR, Francomano CA, and Deng CX (2000) A neonatal lethal mutation in FGFR3 uncouples proliferation and differentiation of growth plate chondrocytes in embryos. *Hum Mol Genet* **9**:1603–1613.
- Jin M, Yu Y, Qi H, Xie Y, Su N, Wang X, Tan Q, Luo F, Zhu Y, Wang Q, et al. (2012) A novel FGFR3-binding peptide inhibits FGFR3 signaling and reverses the lethal phenotype of mice mimicking human thanatophoric dysplasia. *Hum Mol Genet* **21**:5443–5455.
- Kake T, Kitamura H, Adachi Y, Yoshioka T, Watanabe T, Matsushita H, Fujii T, Kondo E, Tachibe T, Kawase Y, et al. (2009) Chronically elevated plasma C-type

- natriuretic peptide level stimulates skeletal growth in transgenic mice. *Am J Physiol Endocrinol Metab* **297**:E1339–E1348.
- Kerr MA and Kenny AJ (1974) The purification and specificity of a neutral endopeptidase from rabbit kidney brush border. *Biochem J* **137**:477–488.
- Krejci P, Masri B, Fontaine V, Mekikian PB, Weis M, Prats H, and Wilcox WR (2005) Interaction of fibroblast growth factor and C-natriuretic peptide signaling in regulation of chondrocyte proliferation and extracellular matrix homeostasis. *J Cell Sci* **118**:5089–5100.
- Kurtz A, Aigner A, Cabal-Manzano RH, Butler RE, Hood DR, Sessions RB, Czubyko F, and Wellstein A (2004) Differential regulation of a fibroblast growth factor-binding protein during skin carcinogenesis and wound healing. *Neoplasia* **6**:595–602.
- Laederich MB and Horton WA (2010) Achondroplasia: pathogenesis and implications for future treatment. *Curr Opin Pediatr* **22**:516–523.
- Long S, Wendt DJ, Bell SM, Taylor TW, Dewavrin JY, and Vellard MC (2012) A novel method for the large-scale production of PG-CNP37, a C-type natriuretic peptide analogue. *J Biotechnol* **164**:196–201.
- Lopez MJ, Garbers DL, and Kuhn M (1997) The guanylyl cyclase-deficient mouse defines differential pathways of natriuretic peptide signaling. *J Biol Chem* **272**:23064–23068.
- Lorget F, Kaci N, Peng J, Benoist-Lasselin C, Mugniery E, Oppeneer T, Wendt DJ, Bell SM, Bullens S, Bunting S, et al. (2012) Evaluation of the therapeutic potential of a CNP analog in a Fgfr3 mouse model recapitulating achondroplasia. *Am J Hum Genet* **91**:1108–1114.
- Lynch SE, Colvin RB, and Antoniadis HN (1989) Growth factors in wound healing. Single and synergistic effects on partial thickness porcine skin wounds. *J Clin Invest* **84**:640–646.
- Maji D, Shaikh S, Solanki D, and Gaurav K (2013) Safety of statins. *Indian J Endocrinol Metab* **17**:636–646.
- Matsushita M, Kitoh H, Ohkawara B, Mishima K, Kaneko H, Ito M, Masuda A, Ishiguro N, and Ohno K (2013) Meclozine facilitates proliferation and differentiation of chondrocytes by attenuating abnormally activated FGFR3 signaling in achondroplasia. *PLoS ONE* **8**:e51569.
- Matsushita T, Wilcox WR, Chan YY, Kawanami A, Bükümmez H, Balmes G, Krejci P, Mekikian PB, Otani K, Yamaura I, et al. (2009) FGFR3 promotes synchondrosis closure and fusion of ossification centers through the MAPK pathway. *Hum Mol Genet* **18**:227–240.
- Modi HN, Suh SW, Song HR, and Yang JH (2008) Lumbar nerve root occupancy in the foramen in achondroplasia: a morphometric analysis. *Clin Orthop Relat Res* **466**:907–913.
- Moncla A, Missirian C, Cacciagli P, Balzamo E, Legeai-Mallet L, Jouve JL, Chabrol B, Le Merrier M, Plessis G, Villard L, et al. (2007) A cluster of translocation breakpoints in 2q37 is associated with overexpression of NPPC in patients with a similar overgrowth phenotype. *Hum Mutat* **28**:1183–1188.
- Murakami S, Balmes G, McKinney S, Zhang Z, Givol D, and de Crombrughe B (2004) Constitutive activation of MEK1 in chondrocytes causes Stat1-independent achondroplasia-like dwarfism and rescues the Fgfr3-deficient mouse phenotype. *Genes Dev* **18**:290–305.
- Murdoch JL, Walker BA, Hall JG, Abbey H, Smith KK, and McKusick VA (1970) Achondroplasia—a genetic and statistical survey. *Ann Hum Genet* **33**:227–244.
- Nakajima H, Ezaki Y, Nagai T, Yoshioka R, and Imokawa G (2012) Epithelial-mesenchymal interaction during UVB-induced up-regulation of neutral endopeptidase. *Biochem J* **443**:297–305.
- Naski MC, Colvin JS, Coffin JD, and Ornitz DM (1998) Repression of hedgehog signaling and BMP4 expression in growth plate cartilage by fibroblast growth factor receptor 3. *Development* **125**:4977–4988.
- Oefner C, D'Arcy A, Hennig M, Winkler FK, and Dale GE (2000) Structure of human neutral endopeptidase (Neprilysin) complexed with phosphoramidon. *J Mol Biol* **296**:341–349.
- Ornitz DM and Marie PJ (2002) FGF signaling pathways in endochondral and intramembranous bone development and human genetic disease. *Genes Dev* **16**:1446–1465.
- Pagel-Langenickel I, Buttgeriet J, Bader M, and Langenickel TH (2007) Natriuretic peptide receptor B signaling in the cardiovascular system: protection from cardiac hypertrophy. *J Mol Med (Berl)* **85**:797–810.
- Peake N, Su N, Ramachandran M, Achan P, Salter DM, Bader DL, Moyes AJ, Hobbs AJ, and Chowdhury TT (2013) Natriuretic peptide receptors regulate cytoprotective effects in a human ex vivo 3D/bioreactor model. *Arthritis Res Ther* **15**:R76.
- Placone J and Hristova K (2012) Direct assessment of the effect of the Gly380Arg achondroplasia mutation on FGFR3 dimerization using quantitative imaging FRET. *PLoS ONE* **7**:e46678.
- Potter LR (2011) Natriuretic peptide metabolism, clearance and degradation. *FEBS J* **278**:1808–1817.
- Potter LR, Abbey-Hosch S, and Dickey DM (2006) Natriuretic peptides, their receptors, and cyclic guanosine monophosphate-dependent signaling functions. *Endocr Rev* **27**:47–72.
- Potter LR and Hunter T (2001) Guanylyl cyclase-linked natriuretic peptide receptors: structure and regulation. *J Biol Chem* **276**:6057–6060.
- Rimoin DL, Hughes GN, Kaufman RL, Rosenthal RE, McAlister WH, and Silberberg R (1970) Endochondral ossification in achondroplastic dwarfism. *N Engl J Med* **283**:728–735.
- Rousseau F, Bonaventure J, Legeai-Mallet L, Pelet A, Rozet JM, Maroteaux P, Le Merrier M, and Munnich A (1994) Mutations in the gene encoding fibroblast growth factor receptor-3 in achondroplasia. *Nature* **371**:252–254.
- Ruchon AF, Marcinkiewicz M, Ellefsen K, Basak A, Aubin J, Crine P, and Boileau G (2000) Cellular localization of neprilysin in mouse bone tissue and putative role in hydrolysis of osteogenic peptides. *J Bone Miner Res* **15**:1266–1274.
- Ryan SM, Mantovani G, Wang X, Haddleton DM, and Brayden DJ (2008) Advances in PEGylation of important biotech molecules: delivery aspects. *Expert Opin Drug Deliv* **5**:371–383.

- Scotland RS, Ahluwalia A, and Hobbs AJ (2005) C-type natriuretic peptide in vascular physiology and disease. *Pharmacol Ther* **105**:85–93.
- Sebastian A, Matsushita T, Kawanami A, Mackem S, Landreth GE, and Murakami S (2011) Genetic inactivation of ERK1 and ERK2 in chondrocytes promotes bone growth and enlarges the spinal canal. *J Orthop Res* **29**:375–379.
- Seymour AA, Mathers PD, Abboa-Offei BE, Asaad MM, and Weber H (1996) Renal and depressor activity of C-natriuretic peptide in conscious monkeys: effects of enzyme inhibitors. *J Cardiovasc Pharmacol* **28**:397–401.
- Shirley ED and Ain MC (2009) Achondroplasia: manifestations and treatment. *J Am Acad Orthop Surg* **17**:231–241.
- Toydemir RM, Brassington AE, Bayrak-Toydemir P, Krakowiak PA, Jorde LB, Whitby FG, Longo N, Viskochil DH, Carey JC, and Bamshad MJ (2006) A novel mutation in FGFR3 causes camptodactyly, tall stature, and hearing loss (CATSHL) syndrome. *Am J Hum Genet* **79**:935–941.
- Trotter TL and Hall JG; American Academy of Pediatrics Committee on Genetics (2005) Health supervision for children with achondroplasia. *Pediatrics* **116**:771–783.
- Veronese FM and Pasut G (2005) PEGylation, successful approach to drug delivery. *Drug Discov Today* **10**:1451–1458.
- Vijayaraghavan J, Scicli AG, Carretero OA, Slaughter C, Moomaw C, and Hersh LB (1990) The hydrolysis of endothelins by neutral endopeptidase 24.11 (enkephalinase). *J Biol Chem* **265**:14150–14155.
- Waldman SD, Usmani Y, Tse MY, and Pang SC (2008) Differential effects of natriuretic peptide stimulation on tissue-engineered cartilage. *Tissue Eng Part A* **14**:441–448.
- Watanabe Y, Nakajima K, Shimamori Y, and Fujimoto Y (1997) Comparison of the hydrolysis of the three types of natriuretic peptides by human kidney neutral endopeptidase 24.11. *Biochem Mol Med* **61**:47–51.
- Wielinga PR, van der Heijden I, Reid G, Beijnen JH, Wijnholds J, and Borst P (2003) Characterization of the MRP4- and MRP5-mediated transport of cyclic nucleotides from intact cells. *J Biol Chem* **278**:17664–17671.
- Wynn J, King TM, Gambello MJ, Waller DK, and Hecht JT (2007) Mortality in achondroplasia study: a 42-year follow-up. *Am J Med Genet A* **143A**:2502–2511.
- Yamashita A, Morioka M, Kishi H, Kimura T, Yahara Y, Okada M, Fujita K, Sawai H, Ikegawa S, and Tsumaki N (2014) Statin treatment rescues FGFR3 skeletal dysplasia phenotypes. *Nature* **513**:507–511.
- Yamashita Y, Takeshige K, Inoue A, Hirose S, Takamori A, and Hagiwara H (2000) Concentration of mRNA for the natriuretic peptide receptor-C in hypertrophic chondrocytes of the fetal mouse tibia. *J Biochem* **127**:177–179.
- Yasoda A, Kitamura H, Fujii T, Kondo E, Murao N, Miura M, Kanamoto N, Komatsu Y, Arai H, and Nakao K (2009) Systemic administration of C-type natriuretic peptide as a novel therapeutic strategy for skeletal dysplasias. *Endocrinology* **150**:3138–3144.
- Yasoda A, Komatsu Y, Chusho H, Miyazawa T, Ozasa A, Miura M, Kurihara T, Rogi T, Tanaka S, Suda M, et al. (2004) Overexpression of CNP in chondrocytes rescues achondroplasia through a MAPK-dependent pathway. *Nat Med* **10**:80–86.
- Yasoda A, Ogawa Y, Suda M, Tamura N, Mori K, Sakuma Y, Chusho H, Shiota K, Tanaka K, and Nakao K (1998) Natriuretic peptide regulation of endochondral ossification. Evidence for possible roles of the C-type natriuretic peptide/guanylyl cyclase-B pathway. *J Biol Chem* **273**:11695–11700.

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